MULTIMODAL COMPUTATIONAL INTELLIGENCE FOR ENHANCING CAREGIVER PERFORMANCE AND CARE DELIVERY

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Individual component: Personalized Sign Language Tutoring System for Caregivers

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Caregivers often look after people who cannot hear or speak, and they are usually the first to see early signs of illness, yet few receive lasting training in either. In a survey of 238 health care providers, only 6.3% had formal sign language training. Existing tools teach general vocabulary, judge signs as right or wrong and ignore the learner's pace. This research aimed to build and evaluate a personalised tutor that helps caregivers keep both skills after training. GLOSS is an individual component of the four-part CareSense platform. It teaches two kinds of non-verbal care signals: 59 care-related American Sign Language signs and Parkinson's motor symptoms. Using design science research, I built a pipeline where MediaPipe landmarks from a webcam clip are classified by an ensemble of a temporal convolutional network and a pose transformer. A recognised sign is graded with dynamic time warping against a reference signer, which tolerates differences in speed. Rule-based feedback names the side and body part to correct, and a large language model rewords it only under a guard that checks every reply. Mastery tracking chooses the next lesson, and a non-diagnostic Parkinson's module uses symptom cards and a video quiz. Selected on validation data only, the ensemble reached a test macro-F1 of 0.7752 (95% CI 0.712 to 0.802) and an accuracy of 0.7859, outperforming the best single model by 0.038 (paired bootstrap 95% CI 0.008 to 0.072). Confident predictions were 98.4% correct, and uncertain attempts prompt a repeat rather than a correction. A full attempt takes 73 to 144 ms, excluding landmark extraction and LLM calls. GLOSS shows that a personalised sign language tutor can recognise care signs reliably, ask for a repeat instead of guessing when unsure, and run on an ordinary webcam and server.

Keywords: sign language tutoring, caregiver training, pose-based sign recognition, dynamic time warping, explainable feedback

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ASL	American Sign Language
BiLSTM	Bidirectional Long Short-Term Memory
CCHSS	Colombo Caregiver and Health Science School
CI	Confidence Interval
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DSR	Design Science Research
DTW	Dynamic Time Warping
FR	Functional Requirement
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
HTTPS	Hypertext Transfer Protocol Secure
IP	Intellectual Property
JSON	JavaScript Object Notation
JSONB	Binary JSON (PostgreSQL data type)
JWT	JSON Web Token
LLM	Large Language Model
LSTM	Long Short-Term Memory
ML	Machine Learning
NFR	Non-Functional Requirement
NVQ	National Vocational Qualification
REST	Representational State Transfer

RGB	Red, Green, Blue
RQ	Research Question
SDG	Sustainable Development Goal
SPOTER	Sign Pose-based Transformer
SUS	System Usability Scale
TCN	Temporal Convolutional Network
WLASL	Word-Level American Sign Language (dataset)
XP	Experience Points

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1 INTRODUCTION

1.1 Overall Project Context

Caregiving is central to both hospital and long-term care, and the demands on caregivers keep growing as populations age. Sustained high workload is linked to burnout and staff turnover, which impose substantial financial costs on health systems [1]. The U.S. Surgeon General has described health worker burnout as a threat to the stability of the health workforce [2]. Patient safety adds a further burden. Falls are common among older adults and are a leading cause of injury and hospitalisation in this group [3]. Caregivers must also document their care activities, and many look after people with hearing loss or limited speech, for whom routine communication is difficult [4]. The group project was motivated by the observation that these problems are usually addressed by separate tools, each designed for a single task and a single type of data.

CareSense is the name of the overall research project (R26-DS-010). It addresses these challenges through an integrated platform made up of four components (PULSE, SENTRY, SCRIBE, GLOSS). Each component was developed by one member of the research group. The components draw on different data modalities (physiological signals, video-derived skeleton data, speech and webcam recordings), but they share one objective: supporting caregiver performance and the quality of care delivered.

- Component 1 PULSE detects caregiver workforce deterioration from physiological and acoustic signals and recommends workload redistribution.
- Component 2 SENTRY predicts fall risk on an edge device from privacy-preserving skeletal movement data.
- Component 3 SCRIBE converts caregiver voice logs in English into structured records of Activities of Daily Living.
- Component 4 GLOSS is a personalised tutoring system that teaches caregivers to interpret non-verbal care signals.

As shown in Figure 1.1, the components share an authentication service, a Supabase database and a single web dashboard. A caregiver can therefore reach all four services through one login and one interface, while each component keeps its own data tables and processing pipeline.

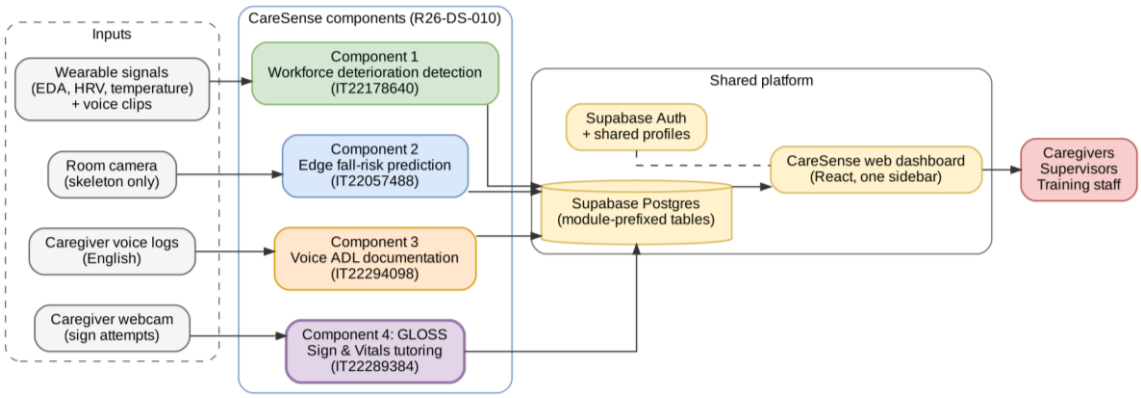


Figure 1.1: Overall architecture of the CareSense platform, showing the four components and the position of GLOSS

1.2 Individual Component Context

GLOSS is the fourth component of CareSense. It addresses the communication side of the caregiver's work: the ability to understand care recipients who cannot rely on speech. Hearing loss becomes markedly more common with age [4], and when a caregiver and a care recipient do not share a language, even basic needs such as water, pain or the toilet become difficult to express.

The focus of the component was shaped by two industry visits. At Remark Elder Care, the chairperson and staff explained that many residents communicate through their bodies rather than through words or signs. Parkinson's disease was identified as one of the common conditions among their elderly residents, and its early motor signs, such as a resting tremor, a shuffling walk or a sudden freeze at a doorway, are easily missed by staff. Parkinson's disease is among the most common neurodegenerative conditions, and its prevalence rises steeply with age [5]. At the Colombo Caregiver and Health Science School (CCHSS) staff described how sign language is introduced in caregiver courses and why it matters for caregivers who later work abroad. Trainees reported that they "only remember a few signs after training".

GLOSS was therefore designed as a tutor for two kinds of non-verbal care signals. The first comes from the person: care-related American Sign Language (ASL) signs, which the caregiver practises in front of a webcam and receives automatic recognition, speed-tolerant execution scoring and explainable feedback on. ASL was selected because it has the largest publicly available video datasets for isolated sign recognition [6], [7]. The second comes from the body: the motor symptoms of Parkinson's disease, taught through symptom cards, short example videos and an adaptive quiz. This second part is strictly educational. It contains no Parkinson's classifier, uses no patient data and makes no diagnostic claim.

1.3 Problem Statement

Health workers know that sign language matters, yet very few have learned it. In a survey of 238 health care providers, 69.3% rated sign language knowledge as very important for communicating with deaf or hard-of-hearing patients, but only 6.3% had received any formal training [8]. The same study found that 59.2% felt uncomfortable communicating with patients who sign and 86.1% wanted training. Deaf signers often receive poorer care because of these gaps [9], and communication problems affect diagnosis and patient satisfaction [10].

Training alone does not close the gap. Signs that are not practised are forgotten, and spaced, repeated practice is what keeps them [11]. Caregivers rarely have a teacher available after their course ends. Self-study tools exist, but most of them teach general vocabulary, judge a sign as simply right or wrong, and say little about what to change [12], [13]. Recognisers trained for benchmarks also tend to treat a slow but correct sign as an error, which discourages beginners who naturally sign slowly.

The Parkinson's side has a similar shape. The motor signs of the disease, such as bradykinesia, rest tremor and rigidity [14], appear gradually and are easy to dismiss as normal ageing. Caregivers see the person every day, so they are well placed to notice change, but only if they know what to look for and know that their job is to observe and report, not to diagnose.

The problem addressed by this research is therefore the following. Caregivers need a way to keep practising care-related signs, and to learn the visible signs of common

conditions, after formal training ends, with feedback that is specific, trustworthy and tolerant of a beginner's pace. No existing tool offers this combination for caregivers.

1.4 Research Gap

The literature review in Chapter 2 compares GLOSS with existing research systems and commercial products. Table 1.1 summarises the gaps identified and how this study responds to each of them.

Table 1.1: Research gaps and the response of this study

Gap	What current work does	Response in GLOSS
Caregiver-specific content	Teaches general ASL vocabulary, or targets emergency services only [12], [15]	59 care signs selected with input from caregiver trainers and reviewed by a sign language tutor
Explanatory feedback	Gives right or wrong judgements; learners cannot tell their own error from a recognition error [12]	Rule-based feedback that names the side and body part to correct, with a hedge when the recogniser is unsure
Separation of speed and accuracy	Only one reviewed prototype compares movements independently of speed (Table 2.1)	DTW comparison against a medoid reference, applied after recognition
Trustworthy AI wording	Uses large language models (LLMs) to write feedback, with limited checking of faithfulness [16]	The LLM may only reword rule text; a guard checks every reply and falls back to the rule text
Rigorous model evaluation	Reports accuracy as a single figure from small evaluations [12]	Selection on validation data only, bootstrap confidence intervals and reproducibility reruns
Personalisation	Adaptive practice is not reported by the reviewed systems (Table 2.1)	Per-sign mastery tracking and an adaptive lesson selector
Non-verbal symptoms	No reviewed system combines sign tutoring with symptom education (Table 2.1)	One tutor covering signs from the person and signs from the body

The research gap is the absence of a caregiver-focused tutor that combines recognition, speed-tolerant evaluation, faithful explanations and adaptive practice and extends the same learning loop to non-verbal symptoms.

1.5 Aim and Objectives

The aim of this research is to design, build and evaluate a personalised tutoring system that helps caregivers learn and retain care-related sign language and recognise Parkinson's motor symptoms, using explainable and adaptive feedback.

The specific objectives are listed below.

- O1. Identify a vocabulary of care-related ASL signs with caregiver trainers and a sign language tutor, and build a clean landmark dataset for it.
- O2. Develop a pose-based sign recogniser, selected on validation data only, and report its test macro-F1 with a 95% confidence interval.
- O3. Implement an execution evaluator that grades how well a recognised sign was performed, independently of signing speed, and identifies the body parts that deviate from the reference.
- O4. Generate interpretable corrective feedback, allow an LLM to reword it only under checks that preserve its meaning, and show the learner a visual guide of the correct sign.
- O5. Track per-sign mastery and select the next lesson adaptively.
- O6. Extend the tutor with a non-diagnostic Parkinson's symptom module.
- O7. Integrate GLOSS into CareSense, evaluate its recognition accuracy, feedback faithfulness and response time, and prepare an evaluation of usability and learning with caregivers.

These objectives follow the proposal, with three changes that are explained in Section 4.3. The proposed real-time 3D avatar was replaced by a 2D landmark guide with a reference video. An optional LLM rewording step, protected by a guard, was added to the rule-based feedback. The Parkinson's module (O6) was added after the visit to Remark Elder Care. The user-based evaluation of learning effectiveness planned in the proposal has been designed, and its execution with caregivers is reported as the next step in Chapter 9.

1.6 Research Questions or Hypotheses

- RQ1. Which pose-based recogniser, selected on validation data alone, recognises the 59 care signs most accurately, and does an ensemble of two models improve significantly on the best single model?
- RQ2. Can the recogniser's confidence be used to decide when the tutor should correct the learner and when it should ask for a repeat?

- RQ3. Can dynamic time warping (DTW) against a reference signer grade execution quality independently of signing speed, and how should its thresholds be calibrated?
- RQ4. Can a large language model reword rule-based feedback without changing its meaning, and which safeguards are needed to ensure this?
- RQ5. How can the tutoring loop be extended to teach non-verbal Parkinson's symptoms while remaining strictly non-diagnostic?

RQ1 is tested through the following hypothesis. H1 states that the averaged ensemble of the improved temporal convolutional network (TCN) and SPOTER-style models achieves a higher test macro-F1 than the best single model, with a paired bootstrap confidence interval for the difference that excludes zero.

1.7 Scope Limitations and Assumptions

GLOSS covers isolated signs, not continuous signed sentences, and teaches American Sign Language only, with instructions in English. The 59 signs come from public ASL datasets, so the reference signers are fluent adults rather than caregivers. The system analyses a short, recorded clip after each attempt and does not yet track signing continuously in real time. The avatar layer proposed as Layer 5 was not completed and is replaced in the product by a 2D landmark guide and a reference video.

The Parkinson's module teaches recognition of six motor symptoms: bradykinesia, tremor, rigidity, postural instability, freezing of gait and shuffling gait. It does not analyse video of any real patient and does not tell the caregiver that anyone has Parkinson's disease. Its message is always to observe, record and report to a nurse or doctor.

The work assumes that learners have a device with a webcam, a modern web browser, adequate lighting and an internet connection. When the camera is not available, the product offers a quiz mode. The evaluation in this thesis covers recognition accuracy, feedback faithfulness and response time. Usability and learning gains with caregivers are outside the scope of the results reported here, and the remaining limitations are discussed in Section 7.5.

1.8 Individual Contribution

GLOSS is my individual component, and I designed, built and evaluated it. Table 1.2 separates my individual work from the shared parts.

Table 1.2: Individual contribution to the GLOSS component

Area	My contribution
Vocabulary and data	Selected care signs with CCHSS input, removed weak classes after checking with a sign language tutor, merged WLASL and ASL Citizen, built the MediaPipe landmark pipeline
Recognition	Trained and compared BiLSTM-Attention, TCN and SPOTER-style models, added kinematic features and augmentation, built the validation-selected ensemble and the single-file product model
Evaluation	Bootstrap confidence intervals, paired gain test, confidence-band analysis, reproducibility reruns, latency tests
Execution scoring	DTW evaluator with medoid references, found and fixed the threshold unit mismatch (v1 to v3)
Feedback	Rule engine, Gemini rewording, faithfulness audit, guard v1 and v2
Learner model	Mastery engine and lesson selector
Parkinson's module	Symptom content design, video quiz, per-symptom progress
Product	FastAPI back end, Supabase schema (migrations 0019 to 0024), React screens inside CareSense
Shared with the team	Common login, sidebar, shared profiles table and repository conventions

1.9 Thesis Structure

Chapter 2 reviews the research and products that GLOSS builds on and develops the identified gaps into a conceptual framework. Chapter 3 explains the research design, the data and the experiments. Chapter 4 sets out the requirements, the architecture and the design decisions, including those that changed during the project. Chapter 5 covers implementation, integration with the other CareSense components and testing. Chapter 6 reports the results, and Chapter 7 discusses their meaning, their comparison with existing work and their limitations. Chapter 8 examines commercialisation and sustainability, and Chapter 9 concludes the thesis and outlines future directions. The appendices contain the evaluation instruments, the AI use disclosure, the product bundle details and the industry visit letters.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The review had two aims: to establish the current state of caregiver-focused sign language tutoring, and to justify each technical choice in GLOSS against published alternatives. Sources were searched in IEEE Xplore, the ACM Digital Library, PubMed Central and Google Scholar. Search terms combined "isolated sign language recognition", "pose", "MediaPipe", "sign language learning", "feedback", "dynamic time warping", "large language model faithfulness", "deaf patients" and "Parkinson's motor symptoms". Most of the work included was published between 2017 and 2026. Earlier foundational works were retained where later research still depends on them, such as the original formulation of DTW and established theories of feedback and tutoring.

A source was included if it reported a working system, an empirical result or a clinical definition on which GLOSS relies. Opinion pieces and studies of continuous, sentence-level sign translation were excluded, because GLOSS addresses isolated signs. Commercial products were reviewed separately through their websites and app store listings, as few of them publish technical papers. Industry input was obtained from the two site visits described in Section 1.2.

2.2 Domain and Theoretical Foundations

2.2.1 Communication with deaf and hard-of-hearing care recipients

The World Health Organization estimates that about 1.5 billion people live with some degree of hearing loss, and that prevalence rises sharply after the age of 60 [4]. For people who sign, the barrier in care settings is linguistic as well as auditory. Jacob et al. argue that deaf signers form a linguistic and cultural group, and that health workers who cannot communicate with them in sign language are less able to provide safe and effective care [9]. Abou-Abdallah and Lamyman show that lip reading, written notes and family interpreters each have limitations, and that these limitations lead to missed information and lower patient satisfaction [10].

The survey by Qadhi provides quantitative evidence of the training gap. Of 238 health care providers, 69.3% considered sign language knowledge very important, yet only 6.3% had received formal training and 86.1% wanted to learn [8]. The most common barrier was a lack of resources (50.0%), followed by a lack of time (21.4%). The survey was conducted in Saudi Arabia, so the exact figures may differ in Sri Lanka, but the pattern of high perceived need and low training is consistent with the findings of the visit. A low-cost tool that caregivers can use in short sessions, at work or at home, addresses both barriers.

2.2.2 Learning motor skills with feedback

Signing is a motor skill, and motor skills improve through practice with feedback. Hattie and Timperley's review shows that feedback is most effective when it tells learners where they are, where they are going and what to do next [17]. A simple correct or incorrect judgement addresses only the first of these questions. Bloom reported that one-to-one tutoring raised achievement by about two standard deviations over conventional classroom teaching [18], and VanLehn's later review found that step-based tutoring systems are nearly as effective as human tutors [19].

Two further ideas shaped GLOSS. Practice that is spaced over time leads to better long-term retention than massed practice [11]. This evidence comes mainly from verbal recall tasks, and its extension to sign vocabulary is an assumption of this work. It explains why the tutor keeps a practice history and brings each sign back over several sessions until it is mastered. Knowledge tracing models a learner's mastery of each skill from their sequence of attempts [20]. GLOSS uses a simple, transparent version of this idea, based on per-sign mastery, streaks and status labels rather than a probabilistic model, because the data available for each learner is limited and the rules need to be explainable to caregivers.

2.2.3 Motor signs of Parkinson's disease

Parkinson's disease is diagnosed clinically. The Movement Disorder Society criteria define parkinsonism as bradykinesia combined with rest tremor, rigidity, or both [14]. Postural instability, freezing of gait and a shuffling walk become common as the disease progresses [21]. The disease also has non-motor symptoms, such as sleep

disturbance and loss of smell, but these are difficult to observe directly and are outside the scope of GLOSS. About 11.9 million people were living with Parkinson's disease in 2021, and prevalence rises steeply with age [5].

The motor signs are observable in everyday activities such as walking, eating and dressing, which places caregivers in a good position to notice change. Early signs, however, are subtle and are easily mistaken for normal ageing. A caregiver who knows that a resting tremor settles when the person reaches for a cup, or that a sudden stop at a doorway may indicate freezing of gait, can give a nurse or doctor a precise description of what was observed. The GLOSS module is designed around this boundary. It teaches observation and reporting, and leaves diagnosis to clinicians.

2.3 Related Research

2.3.1 Pose-based isolated sign recognition

Two public datasets made word-level ASL recognition practical. WLASL collected videos of more than 2,000 signs from online sources and showed that both pose-based and appearance-based models lose accuracy as the vocabulary grows [6]. ASL Citizen added community-sourced recordings made on everyday webcams by documented signers, with signer-independent splits [7]. GLOSS combines both datasets to obtain more signer variety for each of its 59 signs.

Pose estimation reduces the influence of background and lighting. MediaPipe provides on-device pipelines for hands, face and body [22], built on models such as BlazePose [23]. MediaPipe Holistic combines the pose, face and hand models into a single result per frame [24], and GLOSS uses its hand and upper-body landmarks. Working with landmarks rather than pixels also supports privacy, because the video does not need to be stored.

Several model families operate on landmark sequences. Recurrent networks were used as pose-based baselines in the WLASL benchmark [6]. Temporal convolutional networks (TCN) use dilated causal convolutions and outperformed canonical recurrent networks across a range of sequence modelling tasks [25]. SPOTER applies

a transformer encoder [26] to pose sequences and achieved competitive accuracy on WLASL with a lightweight model [27]. Graph convolutional networks model the skeleton as a graph [28], and the multimodal SAM-SLR system, which combined skeleton graphs with RGB streams, won the 2021 large-scale signer-independent isolated sign language recognition challenge [29]. Graph-based and multimodal models are powerful, but they are heavier to train and typically benefit from larger datasets. GLOSS has roughly 28 training sequences per sign. For this reason, GLOSS compared a BiLSTM with attention, a TCN and a SPOTER-style transformer under the same parameter budget.

2.3.2 Ensembles and confidence

Averaging the predictions of different models usually reduces error, because their mistakes do not fully overlap [30]. Deep ensembles also give better-calibrated uncertainty estimates than single networks [31]. This matters for a tutor. Modern neural networks are often over-confident [32], and a tutor that confidently corrects a learner for a sign performed correctly risks undermining the learner's confidence. GLOSS uses ensemble averaging both to improve accuracy and to obtain a confidence value that decides when to correct the learner and when to ask for a repeat.

2.3.3 Speed-tolerant comparison of movements

Dynamic time warping aligns two sequences that unfold at different speeds by stretching the time axis locally [33]. Giorgino's dtw package, of which the dtw-python library used in GLOSS is a port, documents the step patterns and normalisations used in practice, including the symmetric2 pattern [34]. In sign learning, Wen et al. used derivative-based DTW on joint rotations to compare a learner's movements with a teacher's in mixed reality [35]. DTW suits a tutor because it answers a different question from a classifier. A classifier asks which sign was performed. DTW asks how closely the attempt follows a good example of the intended sign, regardless of speed. Its result, however, depends on the choice of reference example, which is why GLOSS uses the medoid of each sign's training examples as the reference.

2.3.4 Feedback for sign language learners

Huenerfauth et al. compared ways of giving feedback on ASL students' videos and found that students preferred feedback shown as pop-ups synchronised with errors in their own video over a separate list of notes, and that both formats helped more than simply reviewing the video [13]. Zhang et al. built a smartphone system for Chinese Sign Language that shows the closest recognised signs, so that learners can compare their attempt with the target [36]. The mixed reality system by Wen et al. scores confusion, smoothness and alignment and returns them as numerical scores for the learner to act on [35].

Large language models have recently been applied in this area. Battisti and Ebling used GPT-4 to convert linguistic annotations of signing errors into feedback that learners can read [16]. This makes feedback more readable, but it raises the question of whether the wording remains faithful to the underlying analysis.

2.3.5 Faithfulness of generated text

A generated text is faithful when it states only what its source supports. Maynez et al. showed that abstractive summarisers frequently add content that is not present in the source [37], and the survey by Ji et al. classifies these hallucinations and the methods used to detect them [38]. For a tutor, an unfaithful rewording is a correctness error. If the rule engine withholds a sign name because the recogniser is uncertain, and the LLM adds it back, the learner receives a confident correction that the evidence does not support. GLOSS therefore treats the LLM as a rewording step under strict checks, rather than as the author of the feedback.

2.3.6 Avatars for sign demonstration

Avatars can demonstrate signs from any angle and at any speed. A recent review by Chemnad and Othman found that avatar intelligibility depends on smooth motion transitions and facial expression, particularly for signers who acquired sign language late, and that abrupt transitions make avatar signing harder to follow [39]. SignAvatar reconstructs 3D sign motion with a learned body model, which illustrates the level of modelling required for natural results [40]. These findings explain why the 3D avatar attempt in GLOSS was replaced, as discussed in Section 4.3.

2.4 Existing Systems Products and Techniques

Table 2.1 compares GLOSS with the closest products and research prototypes. "Not reported" means the feature could not be confirmed from public material.

Table 2.1: Comparison of existing sign learning systems with GLOSS

System	Target users	Webcam assessment	Explains errors	Speed-tolerant scoring	Adaptive practice	Health symptom content
NVIDIA Signs [15]	General ASL learners	Yes	Not reported	Not reported	Not reported	No
Lingvano [41]	General learners, paid app	Not reported	Not reported	Not reported	Not reported	No
Signing for Care [12]	Emergency medical responders	Yes (79 signs)	No; users asked for it	No	No	No
PopSignAI [42]	Novice signers, game	Yes	No	No	Not reported	No
CoSignPlay [43]	Hearing families of deaf children	Avatar-based	No	No	No	No
CSL smartphone tutor [36]	Chinese Sign Language learners	Yes	Shows close matches	No	No	No
Mixed reality tutor [35]	Chinese Sign Language learners	Yes	Scores	Yes (DTW)	No	No
GLOSS (this work)	Caregivers and trainees	Yes (59 care signs)	Yes, side and body part, hedged	Yes (DTW)	Yes	Yes (Parkinson's, non-diagnostic)

Signing for Care is the most closely related system. Its five emergency responders responded positively to the tool, but some could not tell whether a rejected attempt was their own error or a recognition failure, and four of them asked for more detailed feedback to understand and correct their errors [12]. PopSignAI showed that novice signers learned more efficiently when the game asked them to produce signs than in a version where they only recognised them, with an LSTM recogniser that reached 82.9% on an independent test set [42]. NVIDIA Signs offers webcam practice with an avatar for general vocabulary [15]. Lingvano is a paid general course costing USD 19.99 per month [41] and is not aimed at care work.

Apart from the mixed reality prototype, none of these systems separates "which sign" from "how well", and none explains errors at the level of a body part while avoiding corrections when its own recognition is uncertain. None of them treats symptoms of illness as a further kind of non-verbal signal that caregivers need to learn.

2.5 Gap Synthesis and Conceptual Framework

Three findings come out of the review. Feedback has to be specific and tied to the learner's attempt to help a motor skill [17], [13]. A tutor has to know when it does not know, because an over-confident correction is worse than none [32], [12]. And when an LLM is used to make feedback friendlier, its output has to be checked against the evidence [37], [38].

Figure 2.1 turns these findings into the conceptual framework for GLOSS. Signs from the person pass through capture, recognition, execution scoring and explanation. Signs from the body pass through learning, recognition in video and a reporting habit. Both streams feed one learner model, and that model chooses what the caregiver practises next.

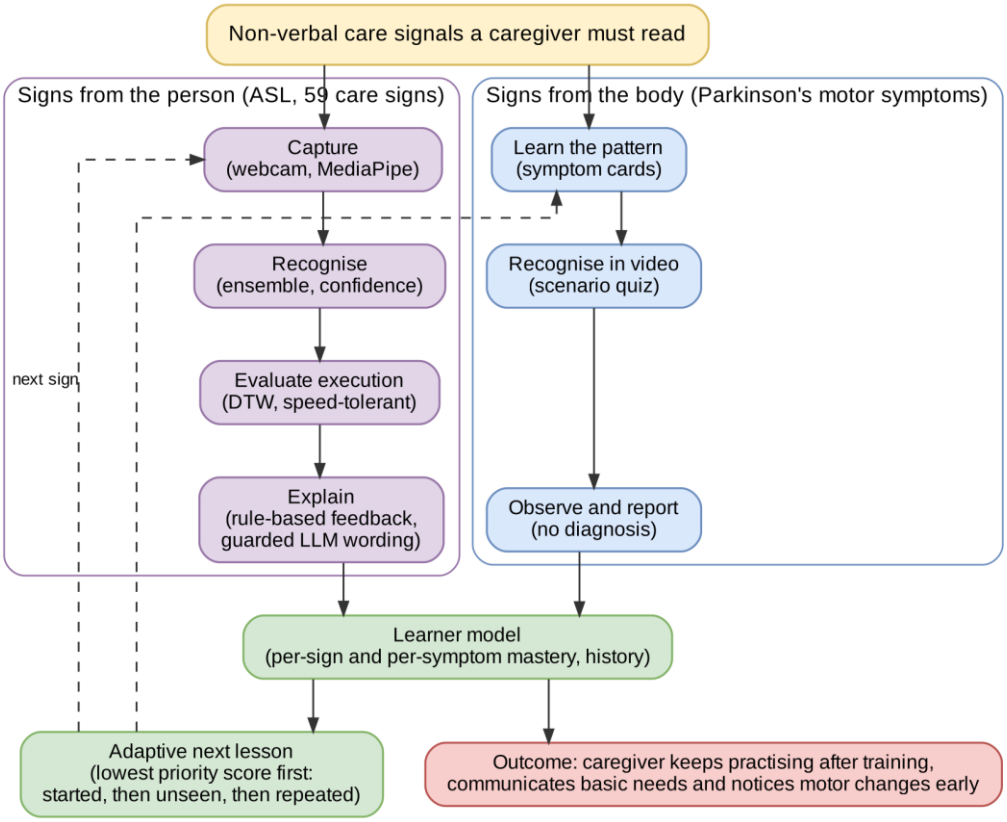


Figure 2.1: Conceptual framework of GLOSS: two kinds of non-verbal care signals feeding one learner model

3 RESEARCH METHODOLOGY

3.1 Research Design

This research follows Design Science Research (DSR), which builds an artefact to solve a real problem and evaluates it rigorously [44]. The process model by Peffers et al. organised the work into six activities [45]. Figure 3.1 shows how they were carried out in GLOSS. The problem was identified from the literature and the two industry visits. Objectives were set in the proposal and revised after the visits. The artefact was then built in layers, demonstrated live to the proposal and progress panels, evaluated with quantitative experiments, and reported in this thesis.

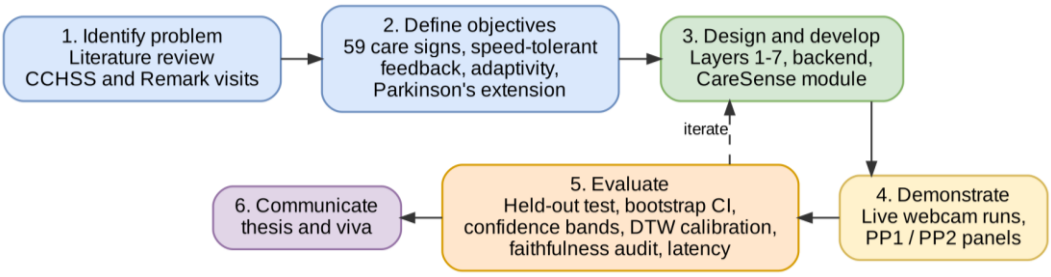


Figure 3.1: Design science research process followed in this study

Inside the "evaluate" activity the study is quantitative and experimental. Recognition models were compared on a fixed split. Each claim about a difference between models is backed by a bootstrap confidence interval. The feedback layer was evaluated with an audit of real LLM outputs. Evaluation fed back into design several times. The DTW thresholds, the choice of recogniser and the feedback guard all changed because an experiment showed a problem.

3.2 Data Participants or Study Materials

3.2.1 Sign vocabulary

The vocabulary started as 66 signs in seven caregiving categories: emergency, health and medical, body parts, greetings, emotions and responses, food and drink, and daily actions. These were chosen from caregiver communication needs discussed at CCHSS. Signs with too few samples in the public datasets, and words rarely used in daily care, were then reviewed with a sign language tutor working in this domain and removed. The final vocabulary has 59 signs, all shown in Figure 6.2.

3.2.2 Video datasets

Videos for the 59 signs were taken from WLASL [6] and ASL Citizen [7]. Using two sources gives more signers per sign and more variety in camera, lighting and background. Table 3.1 summarises the resulting dataset. Three videos failed landmark extraction. An outlier check then found a few unusually long videos in which the signer repeated the sign several times. These were split into individual repetitions, which added 15 sequences and gave the final 2,364.

Table 3.1: Summary of the GLOSS sign dataset

Item	Value
Source videos for the 59 signs	2,352 (WLASL and ASL Citizen)
Videos that failed landmark extraction	3
Sequences extracted successfully	2,349
Additional sequences from outlier rescue	+15
Final landmark sequences	2,364
Landmark extractor	MediaPipe Holistic 0.10.14
Landmarks per frame	49 (both hands and upper body), x, y, z = 147 values
Frames per sequence after resampling	60
Model input (improved models)	60 x 441 (positions, velocity, acceleration)
Split	Stratified 70 / 15 / 15, random_state = 42
Test set	355 sequences, 4 to 10 per sign

The split was stratified by sign so every sign appears in all three parts. It was not split by signer, so the same signer can appear in training and test. This is a known limitation and is discussed in Section 7.5.

3.2.3 Parkinson's symptom content

The Parkinson's module covers six motor symptoms: bradykinesia, tremor, rigidity, postural instability, freezing of gait and shuffling gait. Each symptom has a short description written from clinical sources [14], [21], a "look for" cue telling the caregiver what to watch, and a short memory trick. The quiz bank has 30 questions, of which 12 include a short example video and the rest are text scenarios. No patient data or Parkinson's dataset was used, and no model was trained for this module, as it is an educational extension of the tutoring system.

3.2.4 Industry input

Two site visits informed the requirements. At CCHSS (22 April 2026) the team learned how caregiver courses are structured and how sign language is introduced, and collected the feedback summarised in Section 6.3. At Remark Elder Care (7 April 2026) the chairperson explained the gap between caregiver training and daily practice and recommended teaching non-verbal symptoms. Both organisations issued letters confirming the visits.

3.3 Experimental or Study Procedure

3.3.1 Landmark extraction and preprocessing

Every video was passed through MediaPipe Holistic [24]. Frames where a hand was not detected produced missing values. During data preparation, hands were missing in 23.7% of frames, and forward-fill repair brought this down to 8.2%. Each frame was then normalised so that the signer's position and size in the image do not change the values, and each sequence was resampled to 60 frames. For the improved models, frame-to-frame velocity and acceleration were added to the 147 position values, giving 441 features per frame. Features were standardised with a scaler fitted on the training set only.

3.3.2 Recognition experiments

The recognition experiments ran in three stages, all on the same frozen split.

- Stage A (baselines). A BiLSTM with attention, a TCN [25] and a SPOTER-style transformer [27] were trained on the 147 position features under a similar parameter budget. An earlier set of larger recurrent models had overfitted the small dataset, which is why the budget was imposed.
- Stage B (improvements). The two strongest baselines, TCN and SPOTER-style, were retrained with the 441 kinematic features, data augmentation on the landmark sequences, focal loss [46] and label smoothing [47].
- Stage C (ensemble). The softmax outputs of the improved TCN and the improved SPOTER-style model were averaged [30].

All models used early stopping on the validation set. The final model was chosen by one fixed rule, the highest validation macro-F1. The test set was used only once per model, for reporting. This rule was adopted after repeated training of the single TCN showed large run-to-run differences, and after it became clear that an earlier prototype had been chosen from such runs by its test score. Section 4.3 explains this change.

To check reproducibility, the data preparation notebook and the training of the final ensemble were rerun from scratch, and the outputs were compared with the originals.

3.3.3 Execution quality experiments

For each sign, a reference sequence was chosen as the medoid of its training examples, the example with the smallest total DTW distance to the others. A learner's attempt is compared with this reference using dtw-python with the symmetric2 step pattern [34], and the normalised distance is mapped to a quality tier (strong, moderate or weak) using per-sign thresholds.

The thresholds were calibrated from the distribution of distances between examples of the same sign. Three checks were run. The first compared the calibration distances with the distances produced by the live evaluation function, which uncovered a unit mismatch in the first threshold table (v1). The second measured how much the same-

sign and different-sign distance distributions overlap. The third time-stretched attempts between 0.7 and 1.3 times normal speed and measured how much the normalised distance changed.

3.3.4 Feedback faithfulness experiment

The feedback layer was tested on 59 cases, one per sign. For each case the rule engine produced a message and Gemini [48] produced one rewording. The rewordings were checked in four steps.

1. An automatic keyword checker looked for missing or added content.
2. A stricter manual audit read every rewording and classified problems as serious or minor. A serious problem names a sign the rules had withheld, drops the "not fully certain" hedge, or misreads a sign name.
3. After the guard was built, the 59 cases were rerun with new Gemini calls (guard v1), and every rewording shown to the learner was read manually.
4. A detail-preservation check (guard v2) was applied to the same outputs.

3.3.5 Speed measurement

Inference time was measured in Google Colab over five runs for three set-ups: the single TCN prototype, the ensemble as two separate models, and the single-file ensemble used in the app. A full attempt (recognition, DTW and feedback rules) was also timed. MediaPipe extraction and the Gemini call were excluded because they depend on the device and the network.

3.3.6 Caregiver evaluation protocol

A short user evaluation protocol of 10 to 15 minutes per participant was prepared for caregivers and trainees (Appendix A). Each participant attempts three care signs (help, water and sick) before and after a five-minute practice session with GLOSS, explores the Parkinson's symptom cards and completes one in-app symptom quiz. The session ends with the System Usability Scale (SUS) [49], five GLOSS-specific statements and two questions on training and cost. Signs already known to be weak

for the recogniser were left out of the test so that it measures learning rather than a known model weakness.

3.4 Measures and Evaluation Metrics

Table 3.2 lists the measures used and why each was chosen.

Table 3.2: Evaluation measures

Measure	Definition	Why it was used
Macro-F1	Mean of per-sign F1 scores	Treats rare and common signs equally; the main selection and reporting metric
Accuracy	Share of test sequences recognised correctly	Easy to interpret; reported alongside macro-F1
Per-sign F1	F1 for each sign	Finds weak signs, especially caregiver-critical ones
95% bootstrap CI	Percentile interval over 2,000 test-set resamples [50]	Shows uncertainty on a small test set
Paired bootstrap gain	Difference in macro-F1 between two models on the same resamples	Tests whether the ensemble really beats the best single model
Confidence-band share and accuracy	Share of attempts and accuracy in the bands below 0.35, 0.35 to 0.6, and 0.6 and above	Checks whether confidence can drive feedback decisions
DTW distance and overlap	Normalised DTW distance; overlap of same-sign and different-sign distributions	Checks whether quality grading carries information
Faithfulness	Serious and minor problems per rewording	Checks that the LLM does not change the meaning
Latency	Milliseconds per prediction and per attempt	Checks the interactive requirement
SUS	10-item usability score out of 100 [49], [51]	Standard usability measure for the caregiver evaluation

3.5 Data Analysis

All analysis was done in Python with NumPy, pandas and scikit-learn. Classification reports and confusion matrices came from scikit-learn. For the confidence interval, the 355 test predictions were resampled with replacement 2,000 times and macro-F1 was computed on each resample. The 2.5th and 97.5th percentiles give the interval [50]. For the paired comparison, the ensemble and the best single model were scored on the same resamples, and the interval of the difference was reported together with the share of resamples in which the ensemble was better.

Per-sign F1 values are treated as indicative only. With four to ten test videos per sign, a single video changes a sign's F1 by about 0.2. Confidence bands were fixed at 0.35 and 0.6 before the analysis and were not tuned on the test set. SUS scores are computed with the standard formula and interpreted with the adjective scale of Bangor et al. [51].

3.6 Ethics Privacy Security and Responsible AI

The datasets are public research datasets and were used for research under their published terms. Their licences need to be checked again before any paid deployment, as discussed in Chapter 8.

In the product, the webcam clip is used only to extract landmarks for the current attempt. What the system keeps is the attempt result, such as the recognised sign, confidence, quality tier and feedback, not the video. Caregiver identity is never taken from the client. The back end derives the caregiver profile from the verified Supabase token, so one user cannot write attempts for another.

Responsible AI concerns shaped three design choices. First, below 0.35 confidence the tutor asks for a repeat instead of correcting, because a wrong correction could teach a wrong sign. Second, the LLM only rewords text produced by the rule engine. It never sees videos, the learner's identity or the raw predictions. Third, the Parkinson's module carries a clear non-diagnostic message and teaches caregivers to report observations to clinical staff.

Participants in the caregiver evaluation receive an information sheet, give written consent, are identified only by a code and can stop at any time. The evaluation will be carried out only after approval under the SLIIT research ethics procedure for studies involving human participants.

3.7 Reproducibility and Research Artefacts

The research work is split into three Google Colab notebooks: 01 for data preparation, 02 for model training and selection, and 03 for DTW, feedback and the learner model. Each result in this thesis traces to a CSV or JSON file in a single

evidence folder (GLOSS_Viva_Evidence). Original outputs are kept read-only, and reruns are stored in separate folders so they can be compared.

Two reruns confirmed that the results reproduce. Rerunning notebook 01 rebuilt the same dataset (same videos, labels and sizes), although not bit-identical because files were read in a different order. Retraining the ensemble from scratch gave a test macro-F1 of 0.778 against the original 0.775. The files the product needs are stored in a product bundle with a fingerprint for each file (Appendix C). This lets anyone check that the app runs exactly the model reported here.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Stakeholders

The primary users are caregivers working in care homes or private homes, and caregiver trainees preparing for local or overseas jobs. Supervisors and training coordinators are secondary users who want to see progress. Deaf and hard-of-hearing care recipients and older adults with Parkinson's disease benefit indirectly. Training schools and care home operators are the likely buyers (Chapter 8).

Requirements came from four sources: the proposal, the CCHSS and Remark visits, panel comments at the proposal and progress presentations, and the literature in Chapter 2. They are grouped below as functional and non-functional requirements. Section 6.2 reports how each was met.

4.1.2 Functional requirements

Table 4.1: Functional requirements

ID	Requirement	Source
FR1	Let a caregiver browse the 59 care signs and filter them by practice status	Proposal, CCHSS
FR2	Show a visual guide of each sign (2D landmark animation and reference video) with replay, mirror and slow speeds	Proposal, ASL tutor feedback
FR3	Record a sign attempt with the webcam and send it securely to the back end	Proposal
FR4	Recognise the attempted sign and report a confidence value	Proposal
FR5	Ask for a repeat, without correcting, when confidence is low	Literature, design decision
FR6	Grade execution quality independently of signing speed and locate the deviating body parts	Proposal, ASL tutor feedback
FR7	Give short corrective feedback that names the side and body part to fix	Proposal, reviewer feedback
FR8	Optionally reword feedback with an LLM without changing its meaning	Design decision
FR9	Track mastery, streaks and history for each sign and each caregiver	Proposal
FR10	Recommend the next sign from the learner's practice history	Proposal
FR11	Offer a quiz mode when the camera is not available	Accessibility
FR12	Teach six Parkinson's motor symptoms with cards, example videos and an adaptive quiz, and show per-symptom progress	Remark Elder Care
FR13	Run inside CareSense with the shared login and sidebar	Group integration

4.1.3 Non-functional requirements

The non-functional requirements were framed with the product quality characteristics of ISO/IEC 25010 [52].

Table 4.2: Non-functional requirements

ID	Quality	Requirement	Measure
NFR1	Usability	A caregiver with no technical background completes a practice attempt without help	SUS in the caregiver evaluation
NFR2	Performance	The model path of an attempt stays well under one second on a normal server	Measured latency (Section 6.2)
NFR3	Reliability	Feedback is always shown, even if the LLM fails or is rate-limited	Fallback test
NFR4	Correctness of feedback	The tutor never states a sign or error the evidence does not support	Faithfulness audit
NFR5	Security	Every request is authenticated; the caregiver ID is derived on the server	API tests
NFR6	Privacy	Learner videos are not stored; the LLM receives no personal data	Design review
NFR7	Portability	Runs in a normal browser with a standard webcam, no special hardware	Live demonstrations
NFR8	Maintainability	Model, scaler and thresholds are versioned files with fingerprints	Product bundle manifest
NFR9	Scalability	New signs, learners and modules can be added without redesign	Data design, module boundary

4.2 Component Architecture Within the Complete Solution

Figure 4.1 shows the GLOSS architecture and its external interfaces. It has three tiers.

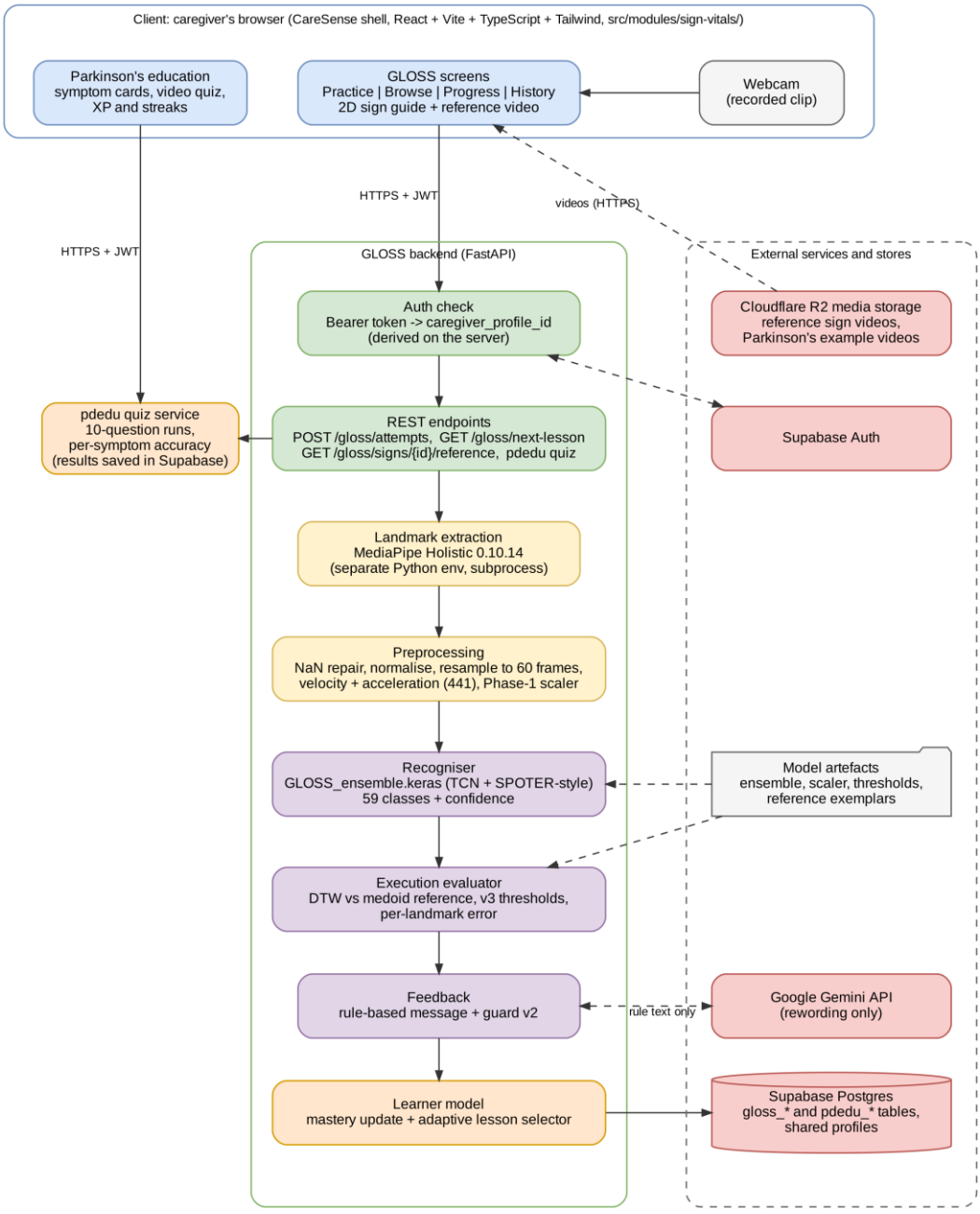


Figure 4.1: GLOSS system architecture showing the client, the back end and the external interfaces

The client tier runs inside the CareSense React application, under the module folder src/modules/sign-vitals/. It provides the sign practice screens and the Parkinson's education screens. The webcam clip is recorded in the browser and sent to the back end with the user's token.

The back end is a FastAPI service. Every request first passes an authentication check that verifies the Supabase token and looks up the caregiver profile on the server. An attempt then moves down the pipeline: landmark extraction, preprocessing, recognition, execution scoring, feedback and the learner model. Landmark extraction runs MediaPipe in a separate Python environment as a subprocess, because the MediaPipe and TensorFlow versions needed by the project conflict in one environment. The Parkinson's quiz is a separate service behind the same endpoints.

The third tier holds the external services and stores. Supabase provides authentication and the Postgres database, and Cloudflare R2 object storage holds the reference sign videos and the Parkinson's example videos. Gemini provides the optional rewording step. The model artefacts are versioned files loaded once at start-up. Table 4.3 lists each external interface and what crosses it.

Table 4.3: External interfaces of GLOSS

Interface	Protocol	Data exchanged	Purpose
Browser to GLOSS API	HTTPS, REST, JSON, Bearer JWT	Video clip, sign ID; returns result, confidence, quality, feedback, mastery, next sign	Practice attempts and lessons
GLOSS API to Supabase Auth	HTTPS	Access token; returns user identity	Authentication
GLOSS API to Supabase Postgres	Supabase client over HTTPS	Sessions, attempts, mastery, sign references, quiz results	Persistence
GLOSS API to MediaPipe	Local subprocess	Video path in; landmark array out	Landmark extraction
GLOSS API to Gemini	HTTPS, API key from environment variable	Rule-based message text only; returns reworded text	Friendlier wording
Browser to Cloudflare R2	HTTPS	Reference sign videos and Parkinson's example videos	Media delivery
GLOSS to CareSense shell	Shared React app, shared auth session	Logged-in user, navigation	Single platform

GLOSS is one of four CareSense components. It shares the login, the profiles table and the dashboard shell with the others, and keeps its own tables (prefixed gloss_ and pdedu_) and its own routes. This keeps each student's component independent while giving users one product. Integration details follow in Section 5.3.

4.3 Design Alternatives and Decisions

Several important decisions changed during the project. They are collected in Table 4.4 and explained below, because the reasons matter as much as the final choice.

Table 4.4: Main design decisions

Decision	Alternatives considered	Choice	Reason
Vocabulary	66 signs in the proposal stage	59 signs	Low-sample and rarely used signs removed after review with a sign language tutor
Recogniser	GRU, CNN+BiLSTM, BiLSTM-Attention, TCN, SPOTER-style, single TCN	Ensemble of improved TCN and SPOTER-style	Best validation macro-F1; statistically supported gain on test; more useful confidence
Model selection rule	Pick the run with the best test score	Highest validation macro-F1 only	Test-set selection is biased; single TCN runs varied a lot
Model packaging	Two models loaded and averaged in the app	One Keras file that averages inside the graph	About half the latency of two predict calls; same outputs
Execution scoring	Rigid frame-by-frame distance; classifier score only	DTW against a medoid reference	Tolerates speed differences; separates "how well" from "which sign"
DTW thresholds	v1 table	v3 table	v1 used a different distance scale, so every attempt looked "strong"
Feedback author	LLM writes feedback from raw predictions	Rules write it; LLM may only reword	Audit found LLM-authored messages added unsupported claims
Visual guidance	3D avatar driven by landmarks	2D landmark guide plus reference video	Lower development cost, clearer display of the landmarks being compared, no 3D loading delay
Back end and store	Flask and SQLite (proposal)	FastAPI and Supabase	Shared CareSense platform, async API, managed auth

4.3.1 From 66 to 59 signs

The proposal-stage dataset covered 66 signs. Some had very few videos in the public datasets, and some were words caregivers rarely use. After a check with a sign language tutor working with this vocabulary, those classes were dropped. The 59 remaining signs still cover the seven categories and include the caregiver-critical signs such as pain, hurt, sick, tired, dizzy, help, doctor, hospital and danger.

4.3.2 Choosing the recogniser

The first experiments used large stacked recurrent models (BiLSTM, GRU and a CNN+BiLSTM hybrid) on 774 features. The hybrid reached 67.18% accuracy on the earlier 66-sign set. These models had far more parameters than the data could support, and their comparison was not fair because their sizes differed. The models were therefore rebuilt under a common parameter budget (Stage A), improved with kinematic features and better losses (Stage B), and finally combined (Stage C).

An early prototype used a single improved TCN. Retraining that TCN with identical settings gave test macro-F1 values anywhere between about 0.66 and 0.78, and the prototype's run had been picked by its test score. That is not a valid way to choose a model. The rule was changed to one fixed criterion, the best validation macro-F1 among all candidates, and that rule chose the ensemble. Section 6.1 shows that the ensemble also has the best test score and that its gain over the best single model holds up under resampling.

4.3.3 Fixing the threshold scale

The first threshold table (v1) was built from DTW distances computed by a different function from the one used during live evaluation. The two differed in scale by roughly a hundred times. The v1 "strong" cut-offs started at 270.7 while real attempt distances were about 3 to 12, so every attempt was graded strong and the grade told the learner nothing. The thresholds were recalibrated with the live evaluation function itself (v2, then v3 with a larger sample). The product uses v3.

4.3.4 Restricting the role of the LLM

The first version of the feedback layer gave Gemini the predicted sign, the confidence and the list of body parts, and asked it to write a friendly message. A manual audit found that 10 of 59 messages had serious problems, such as naming a sign the rules had withheld or dropping the uncertainty hedge (Section 6.1.6). The design was reversed. The rule engine now writes the full message, Gemini sees only that text, and a guard checks every reply.

4.3.5 From 3D avatar to 2D guide

The proposal planned a real-time 3D avatar. A procedural avatar was prototyped in Three.js and React Three Fiber, driven by the reference landmarks, and three factors led to its replacement with a 2D landmark guide. The first was cost. A natural-looking signing avatar requires a rigged and skinned 3D character, motion retargeting and further animation work, which would add substantial development and asset cost to the application [39], [40]. The second was usability. The 2D guide shows exactly the hand and body landmarks that the system compares an attempt against, together with mirror and slow-motion controls, which makes the movement easier for a beginner to follow. The third was loading time. A 3D scene and character must be loaded before a sign can be shown, and this delay was considered a drawback for short practice sessions, whereas the 2D guide appears almost immediately. A reference video of a real signer is offered beside the guide, and a full 3D avatar remains future work.

4.3.6 A lesson from preprocessing

An early preprocessing version normalised every hand point relative to the wrist. Recognition on the original WLASL subset looked excellent, but accuracy on the caregiver vocabulary collapsed to around 5%. Wrist-relative coordinates remove where the hand is in relation to the body, and many care signs differ mainly by location, for example signs made at the head, mouth or stomach. Normalising relative to the body instead fixed the problem. This finding shaped the rule that every preprocessing step must be checked against signs that differ only by location.

4.4 Detailed Design

4.4.1 Layered pipeline

Figure 4.2 shows the seven layers of GLOSS and how a low-confidence result leaves the pipeline early. Layer 5 is drawn dashed because only its 2D replacement is in the product.

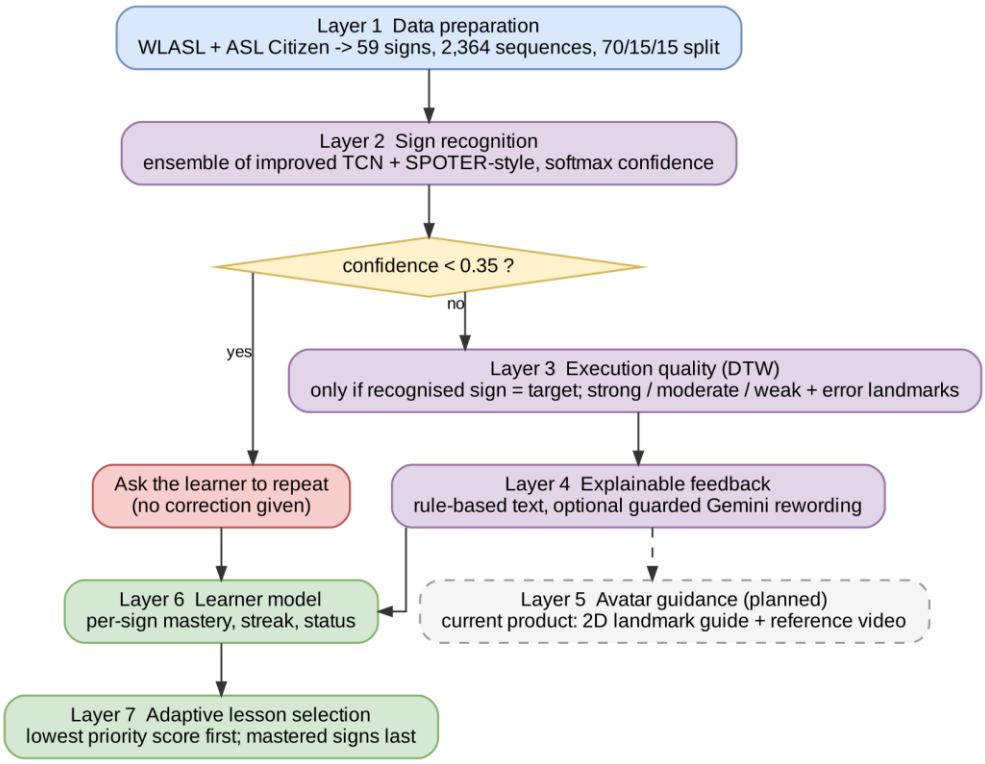


Figure 4.2: The seven GLOSS layers and the low-confidence exit

4.4.2 Recogniser

Figure 4.3 shows the final recogniser. Both members take the same 60 by 441 input. The TCN uses stacks of dilated one-dimensional convolutions with residual connections, so its receptive field covers the whole sign. The SPOTER-style member embeds each frame and passes the sequence through a transformer encoder. Each member ends in a 59-way softmax, and the two probability vectors are averaged. The predicted sign is the class with the highest averaged probability, and that probability is the confidence. The combined model has 202,102 parameters.

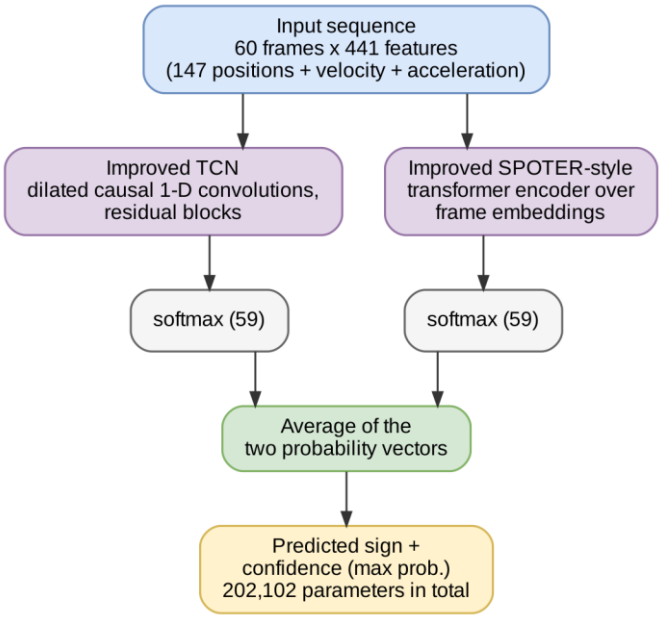


Figure 4.3: Structure of the final ensemble recogniser

4.4.3 Confidence bands

The feedback rules use three confidence bands, fixed by design before evaluation.

- Below 0.35, the tutor does not name a sign and asks the learner to try again.
- From 0.35 to 0.6, the tutor names the likely sign but says it is not fully certain.
- At 0.6 and above, the tutor speaks with confidence.

4.4.4 Execution evaluator

DTW runs only when the recognised sign matches the target sign. Comparing an attempt with the reference of a different sign would produce meaningless corrections. The attempt and the medoid reference are aligned with the symmetric2 step pattern, and the normalised distance is compared with that sign's v3 thresholds to give a tier of strong, moderate or weak. Along the alignment path, the evaluator also measures which landmarks deviate most. These landmarks are mapped to a side and a body part, such as "left wrist" or "right thumb tip", which the feedback rules use. The same medoid sequences drive the 2D guide in the browser, so the learner sees the exact example the attempt is compared against.

4.4.5 Feedback rules and guard

Figure 4.4 shows the feedback flow. The rule engine builds the message from the band, the tier and the deviating landmarks. If confidence is below 0.35, the message is shown as it is and Gemini is never called. Otherwise only the rule text is sent to Gemini with an instruction to reword it without adding facts. The reply must pass every check: no sign name that the rule text did not contain, the hedge kept when the rule text had one, and (in v2) every side and body part kept. If any check fails, or the API returns an error or a rate limit, the learner sees the rule text. The rule text is always correct, so the worst case is a plainer message.

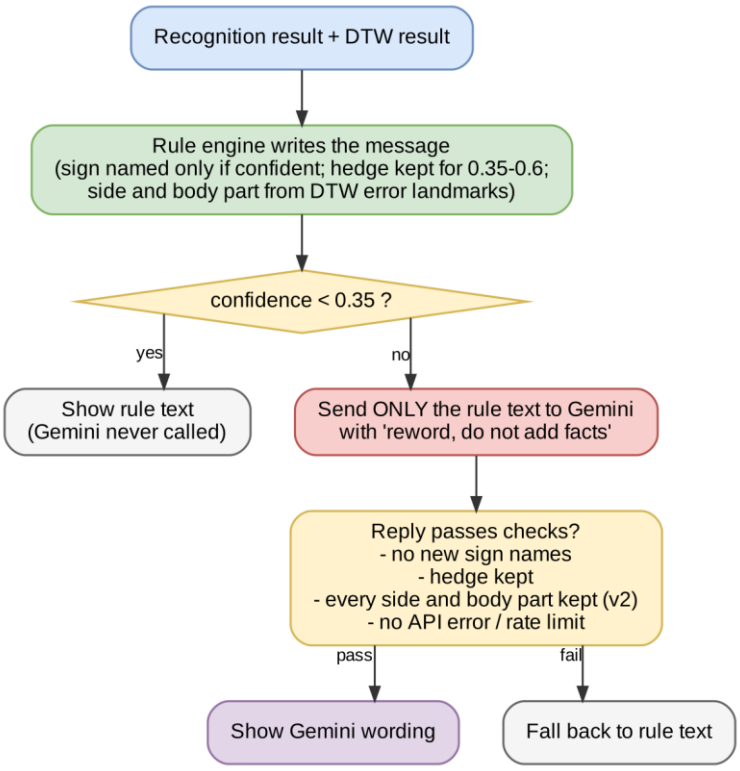


Figure 4.4: Feedback generation with guard v2

4.4.6 Learner model and lesson selection

Each caregiver has one mastery record per sign, holding the number of attempts, the number of recognition mismatches and a streak of consecutive "strong" attempts. When the recognised sign differs from the target, the attempt counts as a mismatch, the streak resets and no quality grade is given. When it matches, a strong attempt

extends the streak and a moderate or weak one resets it. A sign is mastered after three consecutive strong attempts, and a single weaker attempt returns it to practice, so mastery has to be maintained. The interface shows the status as "New", "Learning", "Improving" or "Mastered". The execution score shown after an attempt is for display only and does not affect mastery.

The lesson selector gives every sign a priority score and teaches the lowest first: 5 for a sign never attempted, 100 for a mastered sign, and otherwise max(0, attempts − streak − 3). Signs the learner has recently started are therefore finished before new ones are introduced, and signs that have taken many attempts without progress return later rather than being repeated back to back. Ties are broken at random. The app and notebook versions of these rules gave identical status, streak and score on the same 29-event sequence (tests/test_learner_model.py).

4.4.7 Data design

Figure 4.5 shows the main tables. GLOSS links to the shared CareSense profiles through gloss_caregiver_accounts. For security and management, CareSense has a single administrator account, and only this account can create caregiver accounts, so caregivers cannot register themselves. Each sign has one reference row holding its medoid landmark sequence as JSONB. That row serves both DTW scoring on the server and the 2D guide in the browser. Attempts belong to sessions, and mastery is kept per caregiver and sign. The Parkinson's module stores its symptoms, questions, quiz runs and answers in its own pdedu_ tables. The reference video of each sign and the Parkinson's example videos are stored in Cloudflare R2 and streamed to the browser when needed.

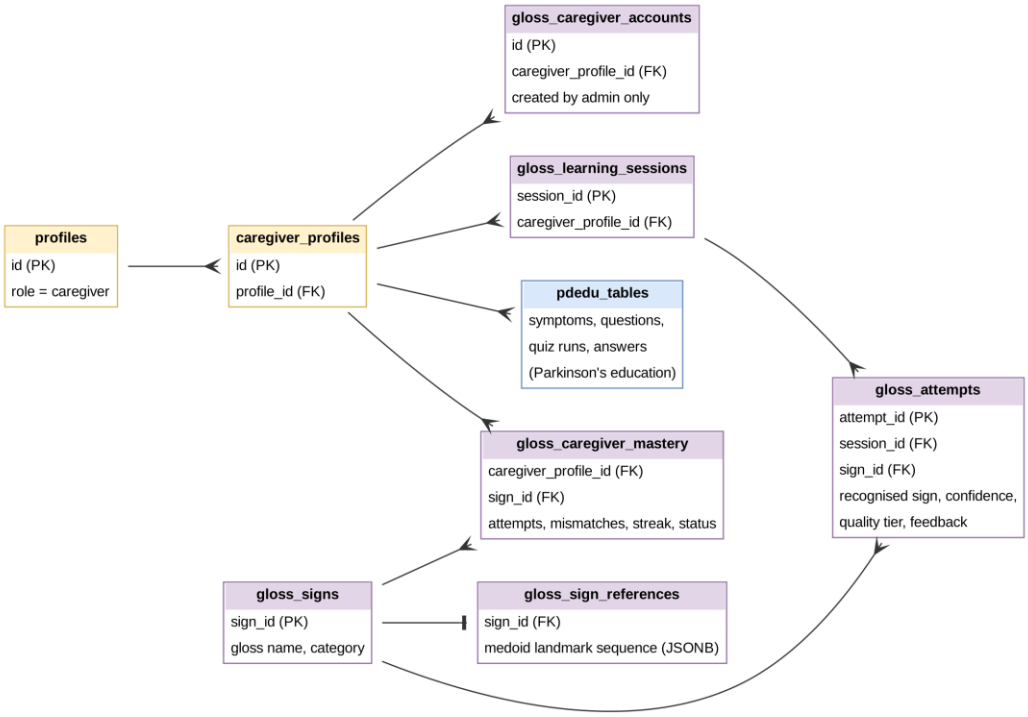


Figure 4.5: Main GLOSS and Parkinson's module tables in Supabase

4.4.8 Parkinson's education module

The module has three parts. The "Explore symptoms" page shows one card per symptom, with a plain description, a "look for" cue, a memory trick and, where available, an example video. The quiz runs ten questions at a time. Video questions ask what is happening in a clip, for example a person who stops walking at a doorway. Each answer updates per-symptom accuracy, and the progress page lists the strongest symptoms and those that need practice. Experience points (XP) and streaks add a light game element. Every screen keeps the same message: notice, record, report.

4.5 Standards and Professional Practice

References follow the IEEE style. The API follows REST conventions over HTTPS with JSON bodies and Bearer JWT authentication. Secrets such as the Gemini key are read from environment variables and are never committed to the repository. Database changes are made through numbered migration files (0019 to 0024 for

GLOSS), so every environment can be rebuilt in order. Code was managed in Git with a shared repository and module boundaries agreed by the team.

Quality requirements were framed with ISO/IEC 25010 [52]. Personal data handling follows three principles: collect only what is needed, keep it secure and use it only for the stated purpose. The Parkinson's module stays within an educational scope, so it does not make the claims that would turn it into a medical device. Third-party libraries are used under their open-source licences.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Research work ran in Google Colab with a GPU runtime and Google Drive for storage. The product was developed locally with Python virtual environments for the back end and Node.js for the front end. Table 5.1 lists the main technologies and why each was chosen over the alternatives.

Table 5.1: Technology stack and justification

Layer	Technology	Alternatives considered	Justification
Landmark extraction	MediaPipe Holistic 0.10.14 [24]	OpenPose	Hands and body in one pass on a CPU, while OpenPose needs a GPU for interactive speed and its licence restricts commercial use; no video needs to be stored; the same version is used in training and in the app so landmarks match
Numerical work	NumPy, pandas, scikit-learn	None	Standard, well-tested tools for preprocessing, metrics and bootstrap analysis
Deep learning	TensorFlow / Keras	PyTorch	The research models were built in Keras, so the saved single-file ensemble runs in the API without conversion; supports TCN and transformer layers
Sequence comparison	dtw-python (symmetric2) [34]	fastdtw	fastdtw approximates the alignment, while dtw-python computes it exactly and documents its step patterns and normalisation
LLM rewording	Google Gemini API [48]	OpenAI GPT models	Either could reword short text; Gemini's free tier covered development, and because the guard only accepts checked replies the provider can be replaced or switched off
Back end	FastAPI (Python)	Flask (proposal)	Same language as the ML code, automatic request validation, async endpoints and generated API documentation
Database and auth	Supabase (Postgres, Auth)	Firebase; SQLite (proposal)	Shared by all CareSense components; a relational schema with numbered SQL migrations, which Firebase's document store does not offer; managed authentication

Layer	Technology	Alternatives considered	Justification
Media storage	Cloudflare R2	Supabase Storage, Amazon S3	Holds the reference video of each sign and the Parkinson's example videos outside the database; R2 charges no egress fees, which matters because the same videos are streamed repeatedly
Front end	React, Vite, TypeScript, Tailwind CSS	None (fixed by the shared shell)	Shared CareSense stack; type checking catches interface errors early
Tooling	Git and GitHub, Graphviz, Colab notebooks	None	Version control, generated architecture diagrams and reproducible experiments

The proposal planned Flask and SQLite. These were replaced because the team agreed on one platform. Supabase gave every component the same login and database, and FastAPI made it easy to validate requests and document the endpoints.

5.2 Component Implementation

5.2.1 Data and model notebooks

Notebook 01 downloads the videos for the 59 signs, runs MediaPipe, repairs missing landmarks, normalises and resamples each sequence, and writes the frozen split. Notebook 02 trains the baselines and the improved models, evaluates them on validation and test, builds the ensemble and runs the bootstrap analysis. It also builds the single-file product model. The ensemble wrapper loads the two trained members and averages their softmax outputs inside one Keras graph. The saved file was checked against the two-model average: the largest difference between the two outputs over the test set was 1.8e-07, and the test macro-F1 was identical (0.7752). Notebook 03 builds the medoid references, calibrates the thresholds, runs the feedback experiments and simulates the learner model.

5.2.2 Back end

The back end mirrors the notebooks. The preprocessing module was written by transcribing the notebook steps exactly, in the same order and with the same constants, so that a live attempt is turned into the same tensor as a training sequence. The model, scaler, thresholds and reference exemplars are loaded once when the

service starts. The main endpoints are POST /gloss/attempts for a practice attempt, GET /gloss/next-lesson for the first recommendation of a session, GET /gloss/signs/{sign_id}/reference for the guide animation, and the quiz endpoints of the Parkinson's module.

The guard is a small, separate module (gloss_feedback_guard.py). Its logic is short enough to show in full as pseudocode in Figure 5.1.

```
def final_feedback(rule_text, confidence, rule_facts):
    if confidence < 0.35:
        return rule_text                        # LLM never called
    try:
        reply = gemini.reword(rule_text)        # LLM sees rule text only
    except (APIError, RateLimit):
        return rule_text
    if mentions_new_sign(reply, rule_facts):  return rule_text
    if rule_facts.hedged and not hedged(reply): return rule_text
    if missing_side_or_body_part(reply, rule_facts):  # v2 check
        return rule_text
    return reply
```

Figure 5.1: Pseudocode of the feedback guard (v2)

5.2.3 Front end

The GLOSS screens live in the shared React application under src/modules/sign-vitals/. Figure 5.2 shows the entry page of the Sign & Vitals area, where a caregiver chooses between the Parkinson's module and GLOSS sign learning.

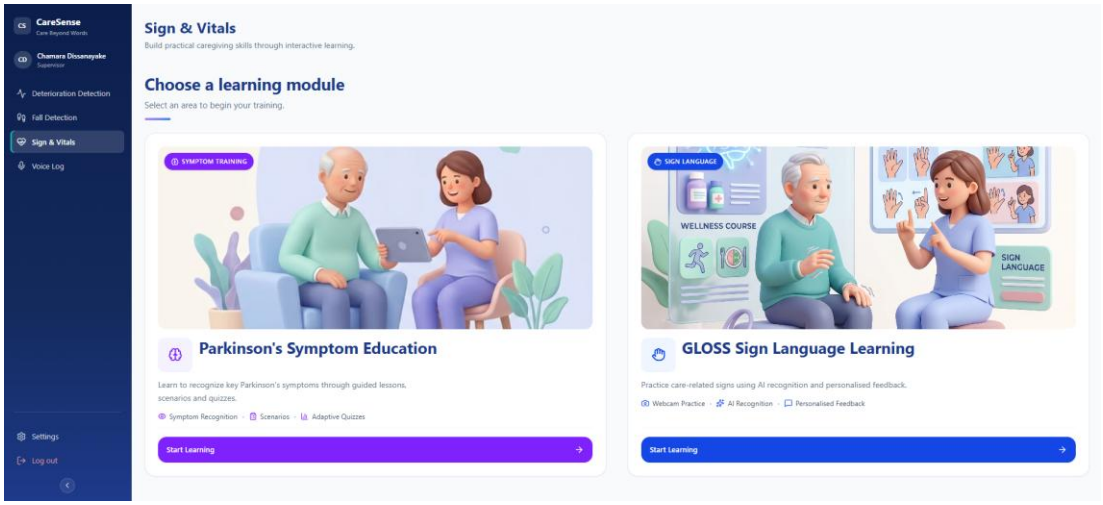


Figure 5.2: Sign & Vitals entry page in CareSense with the two learning modules

The practice home page in Figure 5.3 shows the recommended next sign, its current mastery, and summary cards for signs practised, the practice streak and average

mastery. Recently practised signs can be repeated from the "Continue practising" row.

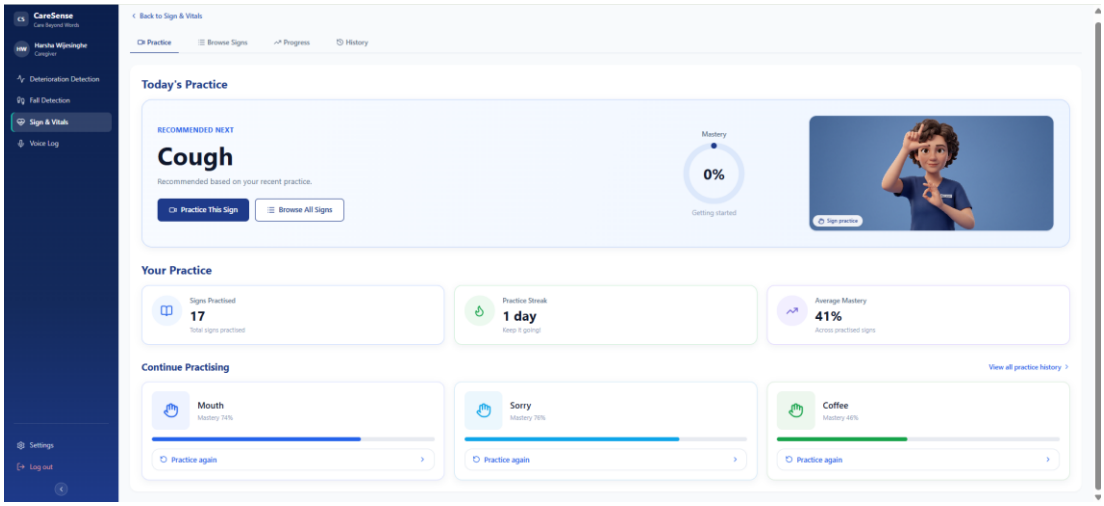


Figure 5.3: Today's practice page with the adaptive recommendation

Before recording, the app asks for camera permission and offers a quiz instead, which covers devices without a camera and learners who prefer not to use one.

During practice, the 2D sign guide plays the reference landmarks of the target sign beside the live webcam preview (Figure 5.4). The learner can pause, replay, mirror the guide and slow it to 0.5 or 0.75 times speed. A tab switches to the reference video. The hint under the preview asks the learner to keep the upper body and both hands in frame with even lighting.

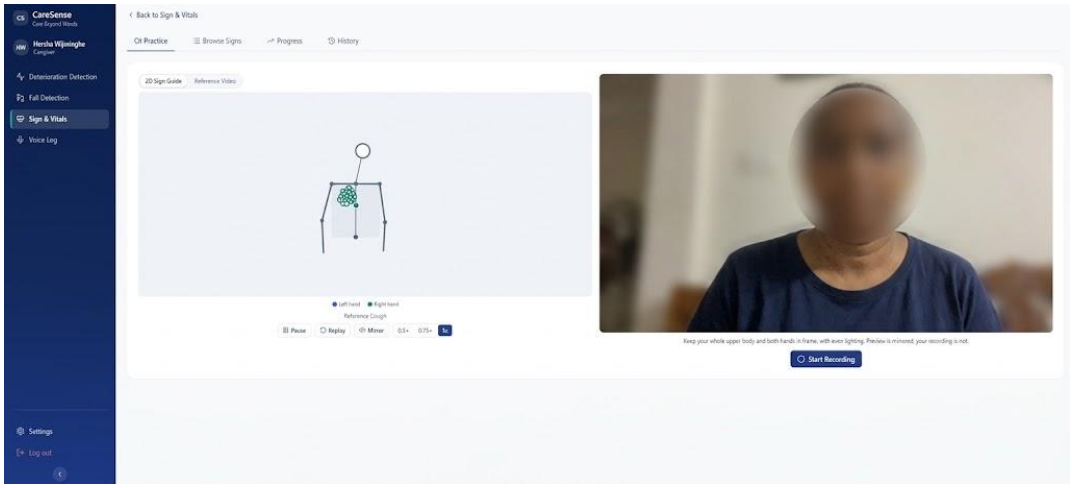


Figure 5.4: Practice screen with the 2D sign guide and the live webcam preview (webcam feed anonymised)

Figure 5.5 shows a real result. The sign "mouth" was recognised correctly with 55% confidence, which falls in the middle band. The message therefore says the system "wasn't fully certain", even though the execution matched the reference closely (quality "strong", execution score 74%). The mastery status, the attempt count and the next actions appear below the message.

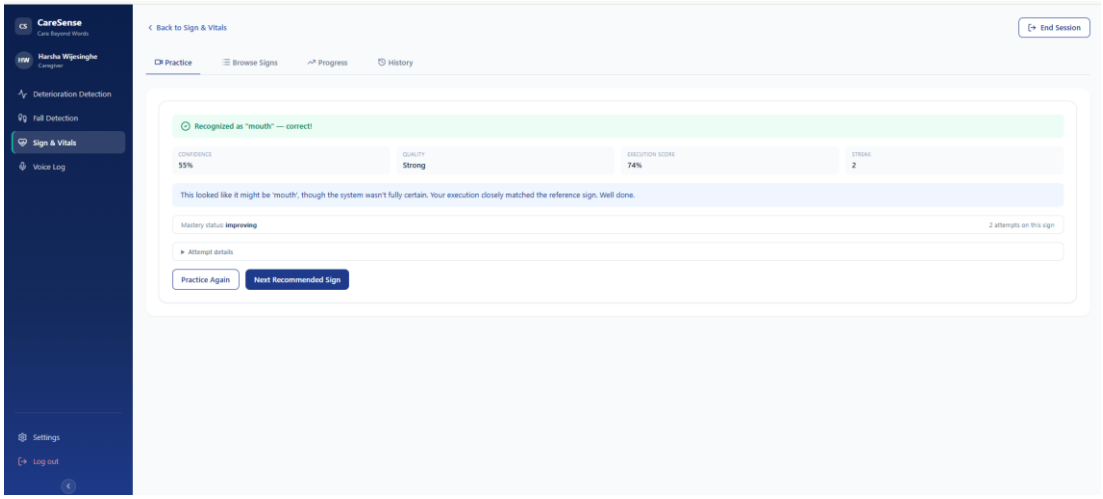


Figure 5.5: Result screen for a real attempt showing the hedged message in the middle confidence band

The other tabs complete the learning loop. "Browse Signs" lists all 59 signs with their status and filters (Figure 5.6). "Progress" shows signs practised, mastered and improving, the total number of attempts, a practice calendar and the most recent attempts (Figure 5.7).

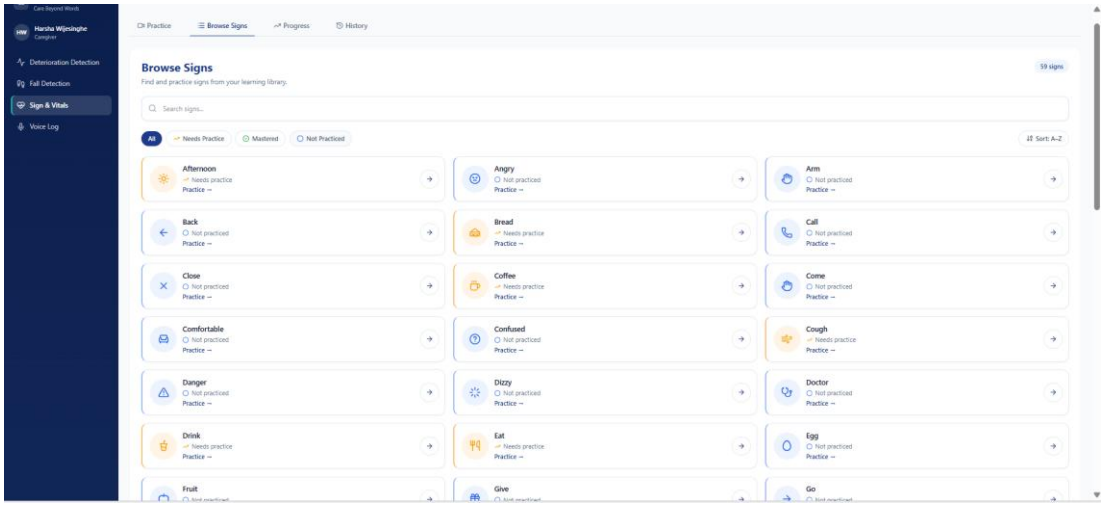


Figure 5.6: Browse Signs page with all 59 signs and practice status

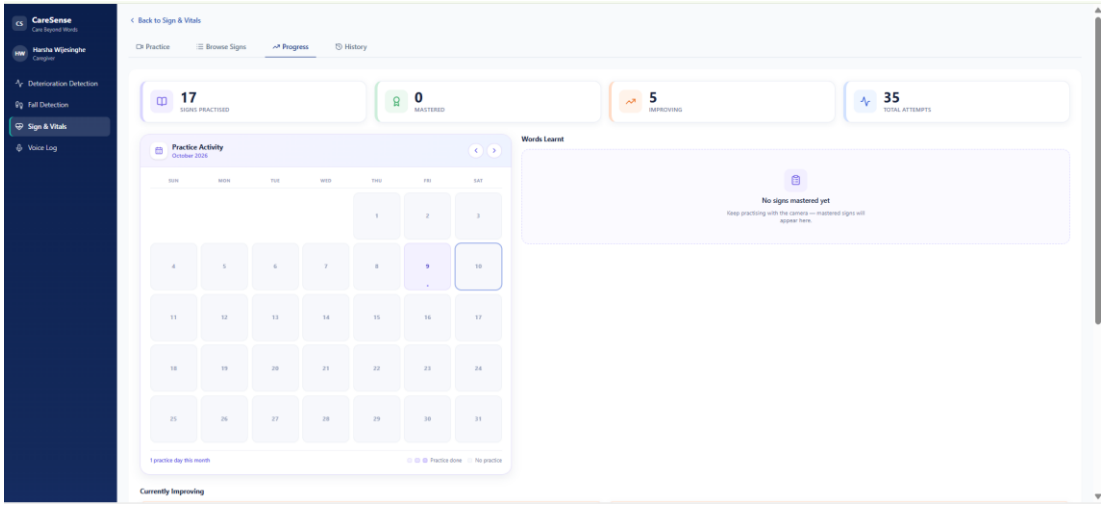


Figure 5.7: Progress page with practice calendar and summary counts

5.2.4 Parkinson's education screens

Figure 5.8 shows the symptom cards, each with a description, a "look for" cue, a memory trick and an example video. The quiz in Figure 5.9 asks what is happening in a short clip. The progress page in Figure 5.10 shows the quiz score, XP, streak and per-symptom accuracy, and lists the strongest symptoms and those that need practice.

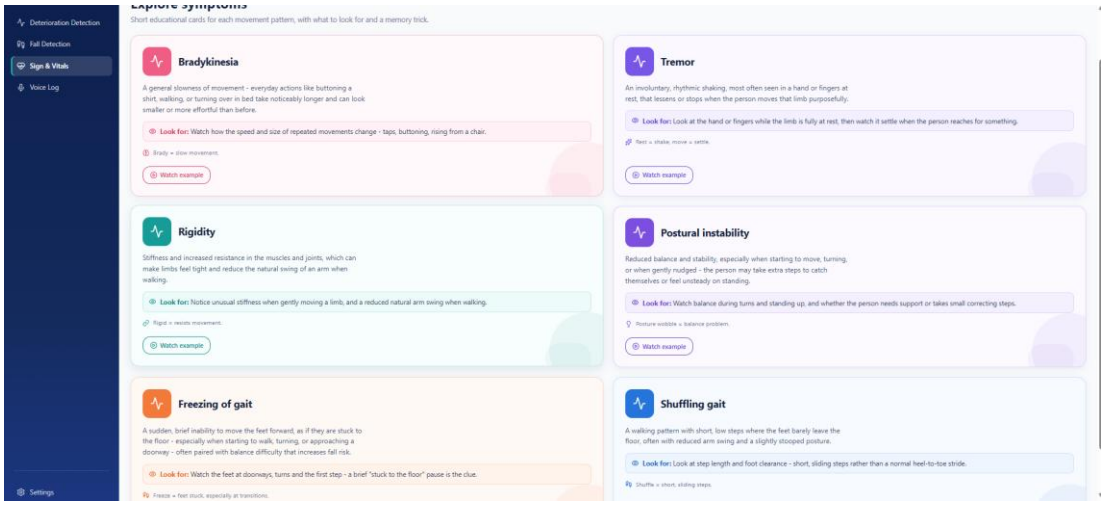


Figure 5.8: Parkinson's symptom cards with "look for" cues

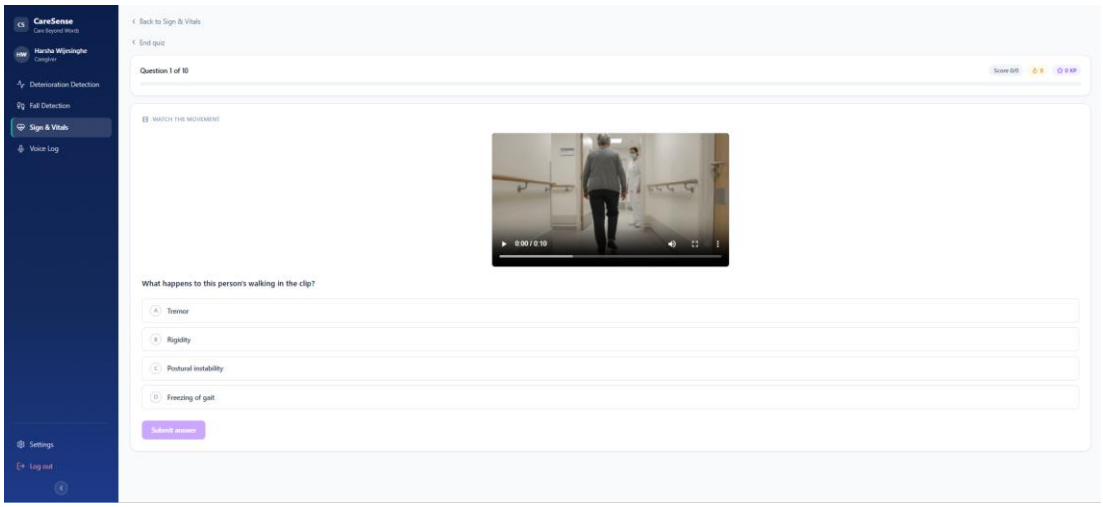


Figure 5.9: Video question in the Parkinson's symptom quiz

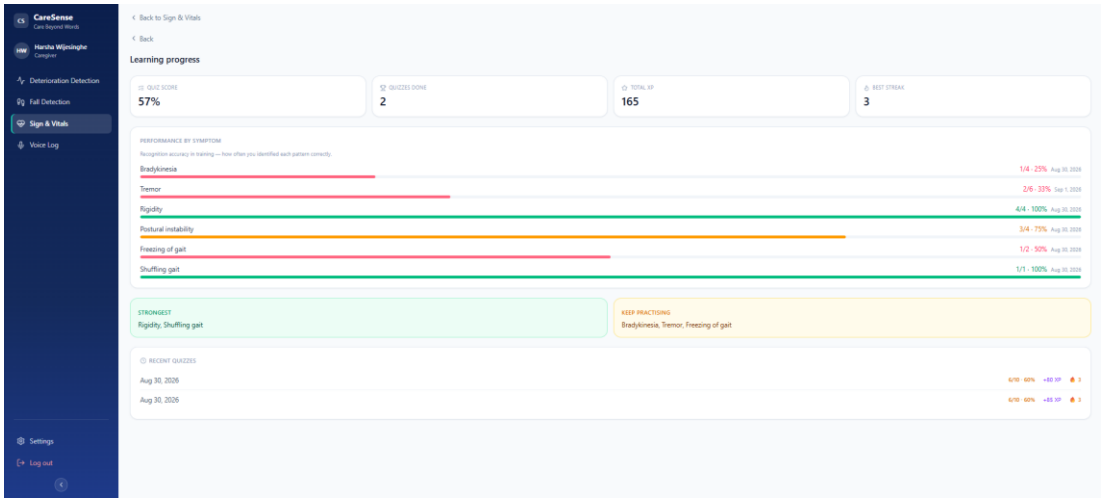


Figure 5.10: Per-symptom learning progress in the Parkinson's module

5.3 Integration with the Other Components

All four components live in one repository (R26-DS-010-Intelligent-Care-Support) and one React application. Figure 5.11 shows how GLOSS fits in. The common shell provides login, the sidebar and settings, and each component owns one sidebar entry and one folder. GLOSS is the "Sign & Vitals" entry.

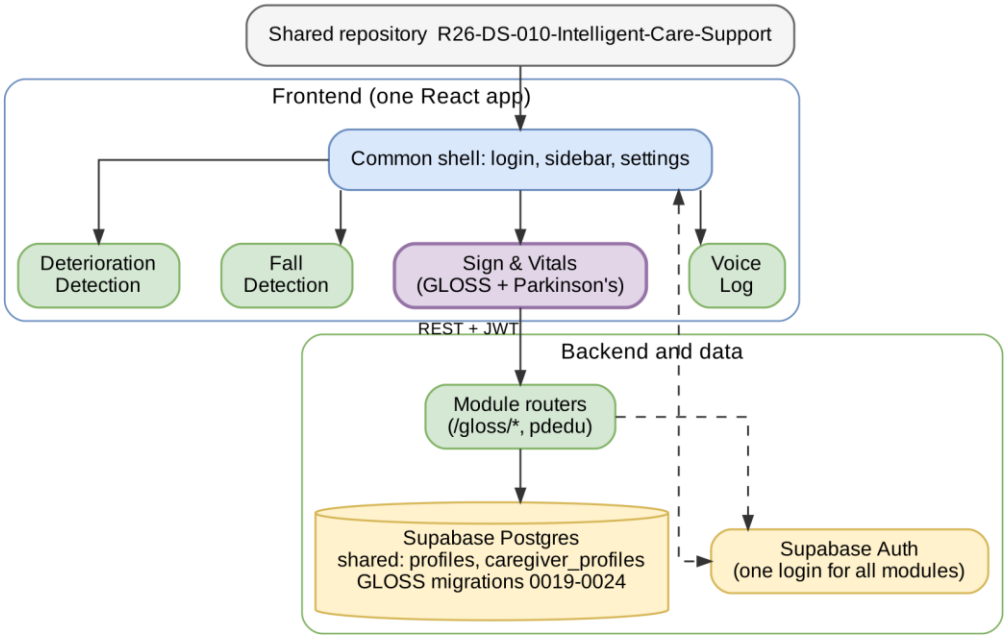


Figure 5.11: Integration of GLOSS into the CareSense repository and platform

Integration needed four agreements.

- Shared identity. CareSense users are rows in a shared profiles table, and caregivers have a role of "caregiver" and a caregiver profile. GLOSS does not create its own users. It links an existing caregiver profile (caregiver_profiles) to GLOSS through gloss_caregiver_accounts. Only the single administrator account can create caregiver accounts, which keeps account management under one point of control.
- Folder and table boundaries. GLOSS code stays in src/modules/sign-vitals/ on the front end and in its own routers on the back end. GLOSS tables carry the gloss_ prefix and the Parkinson's tables the pdedu_ prefix, so no component can break another's data.
- Migration order. The shared database had already reached migration 0018 when GLOSS was added, so the GLOSS migrations were split into one file per table and renumbered 0019 to 0024.
- One login. The front end passes the same Supabase session token to every component's API, and each back end verifies it.

One practical problem came up. MediaPipe 0.10.14 and the TensorFlow version used for the model could not be installed in one Python environment, so landmark extraction runs in its own environment as a subprocess.

5.4 Verification and Software Testing

Testing covered five levels: unit checks of single functions, model parity checks between notebook and product, API tests, end-to-end tests with a live webcam, and non-functional checks. Table 5.2 lists the main test cases and the requirements each one covers. Model and threshold checks (T2, T10 and T12) ran in the notebooks on the frozen test data. T18 and T19 are automated checks in the repository, and T20 to T22 compare the deployed app with stored notebook outputs. The remaining cases ran against the running app and the live Supabase project, with real recorded or live signing rather than mocked data. Several cases test failure and edge conditions on purpose: a missing or forged token (T4, T5), low confidence (T8), a wrong sign (T9), an LLM outage (T11), a rewording that loses detail (T12, T23), a missing camera (T15) and an invalid sign ID (T16).

Table 5.2: Main test cases and results

ID	Test	Input / action	Expected result	Covers	Result
T1	Preprocessing transcription	Recorded test videos through the API pipeline	Same steps and constants as notebook 01; same predictions as the notebook	FR3, FR4	Pass
T2	Single-file model parity	Test set through two-model average and GLOSS_ensemble.keras	Max difference near zero; same macro-F1	FR4, NFR8	Pass (1.8e-07; 0.7752)
T3	Model and scaler load	Start the API	Model, scaler, thresholds and Supabase report ready on /health	NFR8	Pass
T4	Authentication required	Call /gloss/attempts with no token or a bad token	Request rejected, nothing written	FR3, NFR5	Pass
T5	Server-side identity	Valid token with another caregiver's ID in the body	ID in body ignored; profile taken from the token	NFR5	Pass
T6	Correct sign, confident	Live attempt of a clear sign	Sign recognised, quality tier and feedback returned	FR4, FR6, FR7	Pass

ID	Test	Input / action	Expected result	Covers	Result
T7	Middle confidence band	Attempt recognised with confidence 0.35 to 0.6	Hedged message (Figure 5.5)	FR7, NFR4	Pass
T8	Low confidence	Attempt of "pain" recognised as "walk" at 0.23	Repeat request, no correction	FR5, NFR4	Pass
T9	Wrong sign	Different sign performed	"Not the target" message; DTW not run	FR6, NFR4	Pass
T10	Threshold scale	Twelve attempts graded with v1 and v3	v3 gives a spread of tiers	FR6	Pass (v1: 12 strong; v3: 2 / 4 / 6)
T11	LLM failure	Gemini unavailable or rate-limited	Rule text shown, no error to user	FR8, NFR3	Pass
T12	Guard v2 detail check	Rewording that drops "left" from "left wrist"	Fallback to rule text	FR8, NFR4	Pass
T13	Mastery update	Several attempts on one sign	Streak and status change; history updated	FR9	Pass
T14	Next lesson	New caregiver with no attempts	A recommended sign is returned	FR10	Pass
T15	Camera denied	Choose "Use quiz instead"	Quiz mode opens	FR11	Pass
T16	Reference endpoint	Request the reference of a valid and an invalid sign	Landmark sequence; clear error for invalid ID	FR2	Pass
T17	Parkinson's quiz run	Complete a 10-question quiz	Score, XP and per-symptom accuracy saved and shown	FR12	Pass
T18	Front-end build	Type check, lint and production build	No errors	FR13, NFR7	Pass
T19	Learner-model parity	The same 29 attempt events fed to the notebook rules and the app (tests/test_learner_model.py)	Identical status, streak and priority score after every event	FR9, FR10	Pass (29 / 29)
T20	App model parity	Ten stored test sequences (parity_check.npz) through the deployed app	Same predicted sign and probabilities as the notebook	FR4, NFR8	Pass (10 / 10; max diff 2.4e-07)
T21	Full-pipeline parity	Ten stored complete attempts (pipeline_parity.npz) through the deployed app	Same sign, confidence, DTW distance, tier, landmarks and feedback text	FR4, FR6, FR7	Pass (10 / 10)
T22	Execution score mapping	Strong, moderate and weak attempts	Displayed score never contradicts the quality tier	FR6	Pass

ID	Test	Input / action	Expected result	Covers	Result
T23	Live guard check	Six real Gemini calls (strong, moderate and weak attempts, two runs)	Faithful replies accepted; a reply that loses a detail rejected	FR8, NFR4	Pass (5 accepted, 1 rejected to rule text)

The parity files deserve a note. The first product build ran the single TCN prototype, and the screenshots in Section 5.2 were taken with it. The product bundle now carries the ensemble and guard v2 as the release version, together with ten test sequences and their expected outputs (parity_check.npz) and ten complete attempts (pipeline_parity.npz). The deployed app reproduced both sets exactly (T20 and T21), which shows that it runs the model reported in this thesis. FR1 (browsing and filtering signs) and the replay, mirror and speed controls of FR2 are shown working in Figure 5.6 and Figure 5.4 and were checked by inspection rather than by a scripted test. Any later deployment must pass the parity checks before it goes live.

Non-functional testing measured latency (Section 6.2), checked that the guard keeps feedback available when the LLM fails, and reviewed the privacy design. Live end-to-end tests with a webcam were repeated many times during development and in front of the evaluation panel at the second progress presentation.

5.5 Deployment and Operation

For development and demonstrations, the front end runs with Vite and the back end with Uvicorn on a laptop, connected to the shared Supabase project. For a pilot, the plan is a static front-end deployment, the FastAPI back end in a container with at least 2 GB of memory (MediaPipe and TensorFlow together need it), and a Supabase Pro project. The Gemini key and the Supabase service key are supplied as environment variables. Operating GLOSS needs only a few routine tasks. The administrator account creates caregiver accounts, model files are replaced as a set with their manifest, and the parity test runs after each deployment. One limitation was found during code review. The attempt handler is asynchronous, but the Supabase client, the model call and the MediaPipe subprocess are blocking calls. A single attempt therefore holds the server for its full duration. Moving these calls to a worker pool is the first change planned before a multi-user pilot.

6 RESULTS AND VALIDATION

6.1 Research Results

6.1.1 Model comparison

Table 6.1 and Figure 6.1 show validation and test macro-F1 for every candidate. Validation scores are higher than test scores because validation data was used for early stopping. The test score is the reported result throughout.

Table 6.1: Recognition results on the 59-sign task (test n = 355)

Stage	Model	Validation macro-F1	Test macro-F1
Baseline (147 features)	BiLSTM-Attention	0.6065	0.5994
Baseline (147 features)	TCN	0.7196	0.6397
Baseline (147 features)	SPOTER-style	0.7299	0.6890
Improved (441 features)	TCN	0.7437	0.6622
Improved (441 features)	SPOTER-style	0.8031	0.7376
Final	Ensemble (TCN + SPOTER-style)	0.8116	0.7752

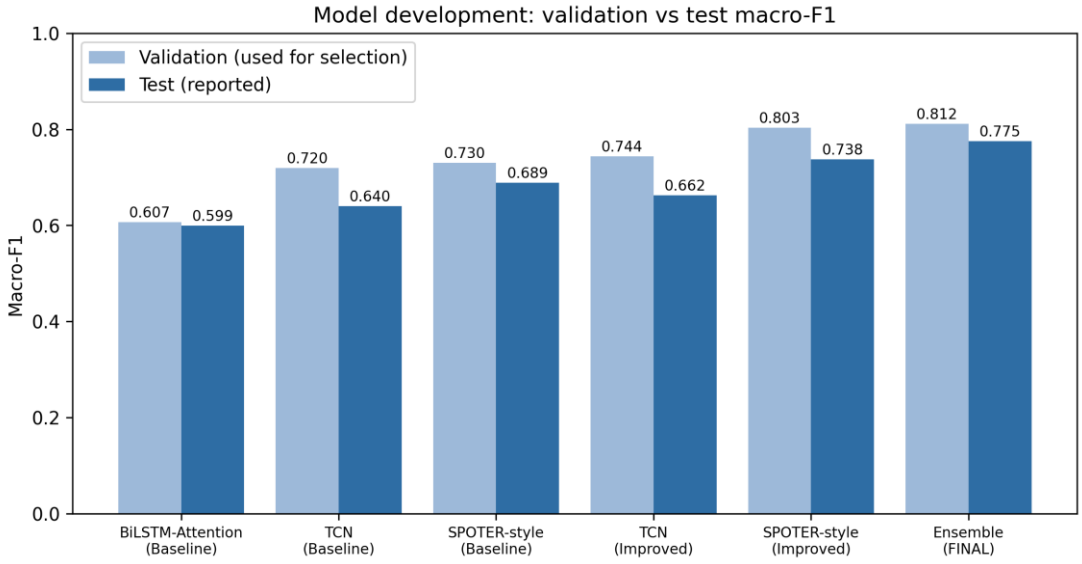


Figure 6.1: Validation and test macro-F1 of all candidate models

Three patterns stand out. The BiLSTM with attention was the weakest model and was dropped after Stage A. The kinematic features, augmentation and loss changes helped both remaining models, and SPOTER-style gained most (0.689 to 0.738). The

ensemble ranked first on validation, which is the only criterion used for selection, and it also has the best test score.

6.1.2 Final recogniser and the ensemble gain

On the 355 test videos, the ensemble reached a macro-F1 of 0.7752 (95% CI 0.712 to 0.802) and an accuracy of 0.7859. Its test macro-F1 was 0.038 higher than that of the best single model, the improved SPOTER-style (0.7752 against 0.7376). In 2,000 paired bootstrap resamples, the mean gain was 0.039 (95% CI 0.008 to 0.072), and the ensemble scored higher in 99.5% of resamples. The interval excludes zero, so hypothesis H1 is supported. Retraining the ensemble from scratch gave a test macro-F1 of 0.778, close to the original 0.775.

6.1.3 Per-sign results

Figure 6.2 shows the F1 of each sign. Six signs reached 1.00: happy, hungry, nose, open, scared and tea. Most signs lie between 0.7 and 0.95. Table 6.2 lists the weakest signs.

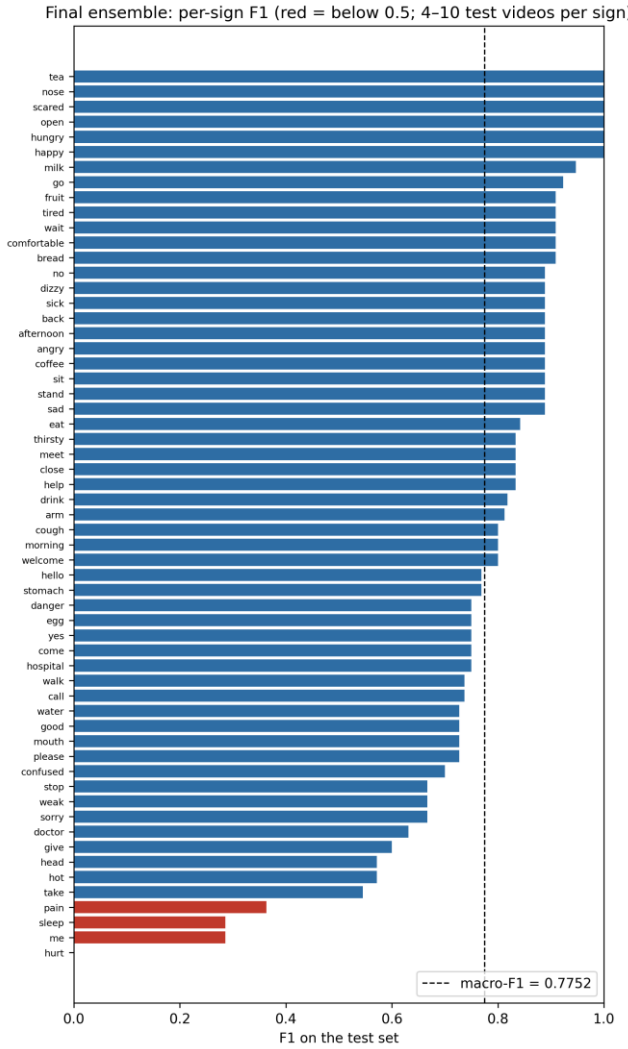


Figure 6.2: Per-sign F1 of the final ensemble on the test set

Table 6.2: Weakest signs of the final ensemble

Sign	F1	Test samples
hurt	0.00	4
me	0.29	5
sleep	0.29	5
pain	0.36	5
take	0.55	5
head	0.57	5
hot	0.57	5

Two of the weak signs, hurt and pain, are among the most important for caregivers. Figure 6.3 shows the confusion matrix for the caregiver-critical signs. All four "hurt" videos were predicted as "pain", and two of the five "pain" videos were predicted as "hurt". In ASL these two signs share a very similar handshape and movement, so the

confusion is understandable, but it matters in care. Other critical signs did well: sick 4 of 5 (F1 0.89) and tired 5 of 5 (F1 0.91). Confused reached 7 of 10 (F1 0.70).

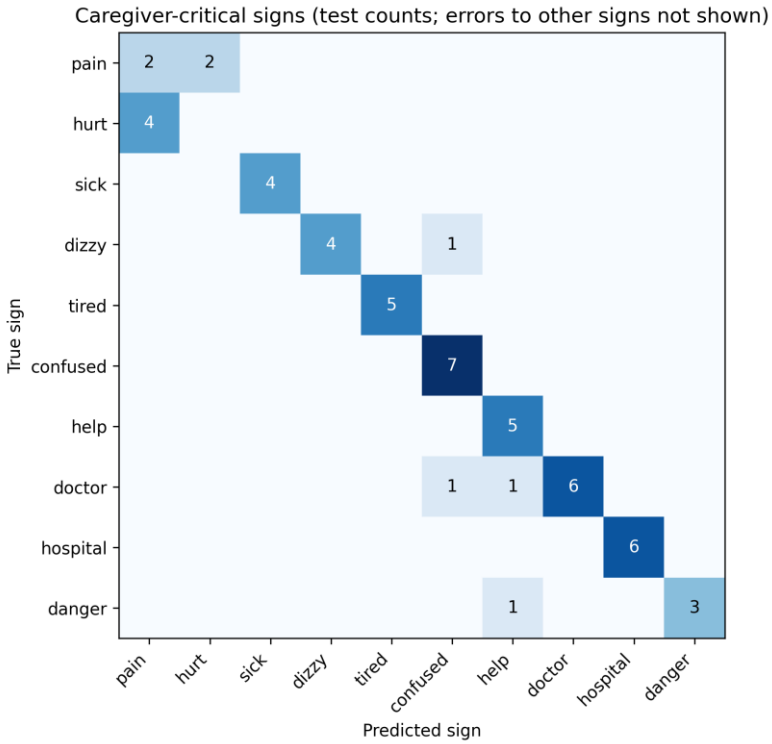


Figure 6.3: Confusions among caregiver-critical signs on the test set

The full row-normalised confusion matrix in Figure 6.4 shows that most errors are scattered single videos rather than systematic pairs. The main exceptions are hurt to pain and sleep, which is spread across several signs. With four to ten test videos per sign, one video moves a sign's F1 by about 0.2, so per-sign numbers should be read as indicative.

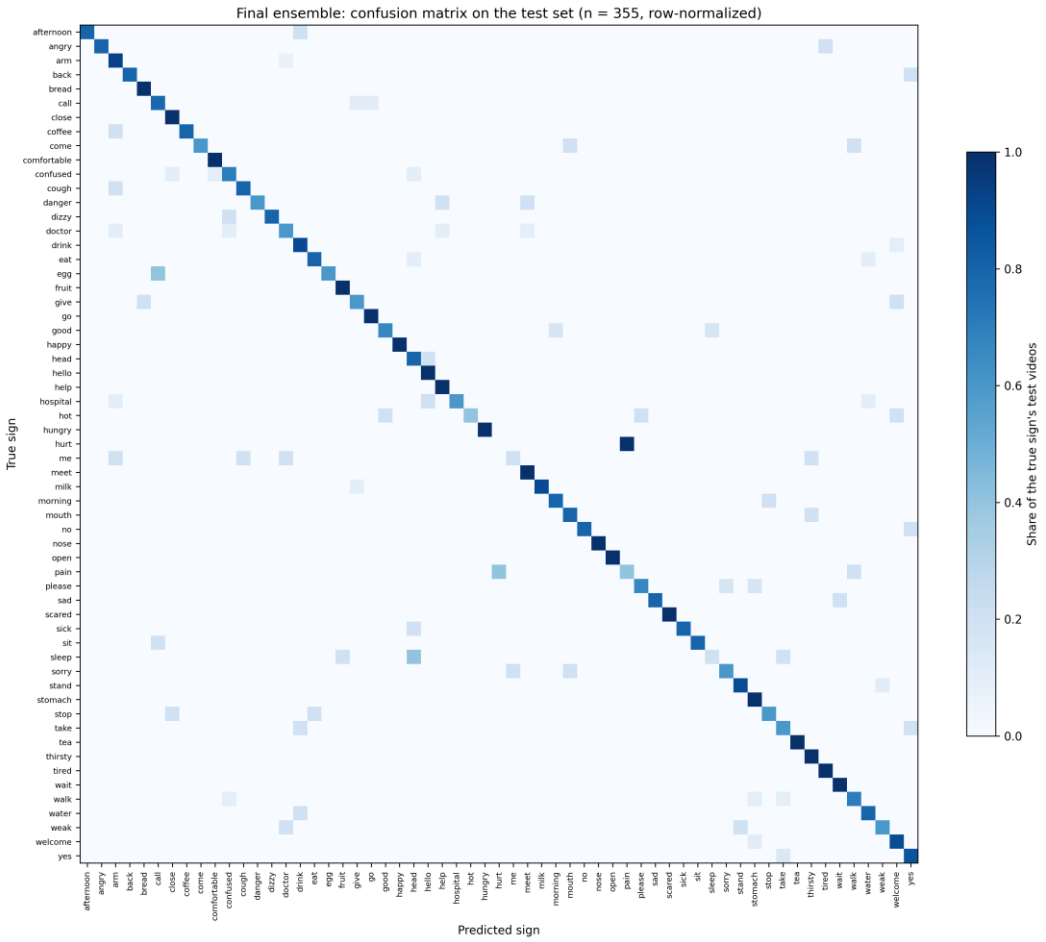


Figure 6.4: Row-normalised confusion matrix of the final ensemble (n = 355)

6.1.4 Confidence as a decision signal

Table 6.3 and Figure 6.5 compare how the single TCN prototype and the final ensemble spread their test predictions across the three confidence bands.

Table 6.3: Share of test attempts and accuracy in each confidence band

Band	Single TCN share	Single TCN accuracy	Ensemble share	Ensemble accuracy
Below 0.35 (ask to repeat)	30.4%	50.9%	22.3%	41.8%
0.35 to 0.6 (not fully certain)	38.9%	77.5%	42.3%	81.3%
0.6 and above (confident)	30.7%	98.2%	35.5%	98.4%

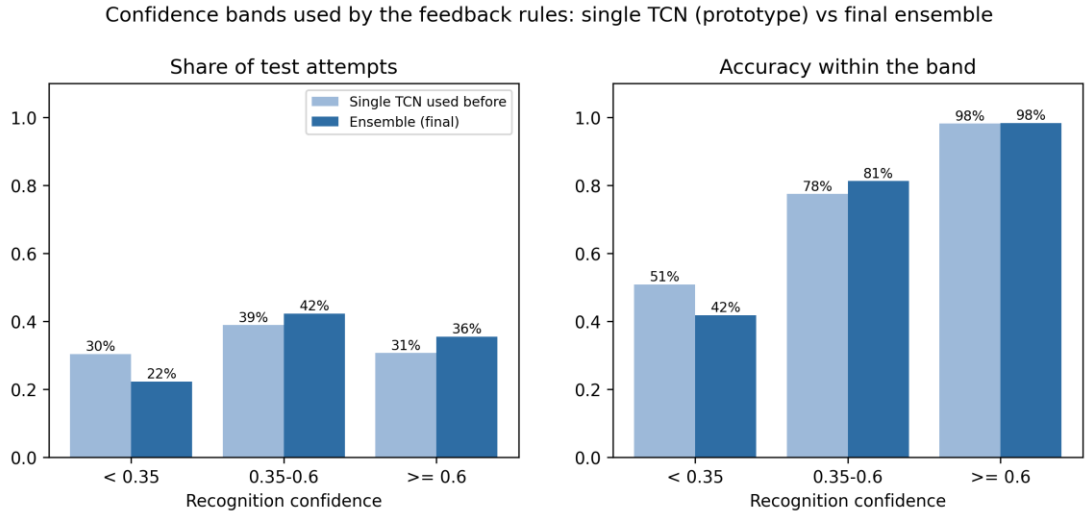


Figure 6.5: Confidence bands used by the feedback rules: single TCN prototype and final ensemble

When the ensemble is confident it is right 98.4% of the time, and this band covers 35.5% of attempts. The ensemble asks for a repeat on 22.3% of attempts, against 30.4% for the single TCN, so learners are interrupted less often. The cost is also clear. Of the attempts the ensemble asks to repeat, 41.8% had in fact been recognised correctly. GLOSS accepts this trade-off on purpose, because asking for one more attempt is far less harmful than correcting a learner who signed correctly. The 0.35 and 0.6 cut-offs were fixed before this analysis and were not tuned on the test set.

6.1.5 Execution quality and speed tolerance

After recalibration, the mean distance between a sign and its own reference was 5.60 with a standard deviation of 3.87 (n = 708). The overlap between the same-sign and different-sign distance distributions fell from 69% with the first calibration to 50.85% with the corrected one. That is a real improvement, but it also means DTW alone cannot tell every pair of signs apart. Some pairs, such as come, call, good, happy and walk, are hardly separable by DTW distance. This is why GLOSS uses DTW only after the recogniser has confirmed the sign. Its job is to grade how well the sign was made, not to decide which sign it was. Table 6.4 shows the effect of the threshold fix on the same 12 sampled attempts.

Table 6.4: Quality tiers for 12 sampled attempts with the v1 and v3 thresholds

Threshold table	Strong	Moderate	Weak
v1 (strong cut-offs of 270.7 and above)	12	0	0
v3 (cut-offs 2.23 to 9.70)	2	4	6

With v1 every attempt was "strong", so the tier gave the learner no information. With v3 the tiers spread out as expected.

Speed tolerance was checked by replaying attempts at 0.7 to 1.3 times normal speed. Across this range, the normalised DTW distance changed by 0.337. That is small next to the spread of distances within a single sign (standard deviation 3.87), so signing 30% slower or faster has little effect on the grade compared with how the sign is actually formed.

6.1.6 Faithfulness of the feedback wording

Table 6.5 summarises the feedback experiment on 59 cases.

Table 6.5: Faithfulness of LLM rewording (n = 59, one rewording per sign)

Check	Result
Original automatic keyword checker	54 / 59 passed
Keyword checker with false alarms fixed	59 / 59 passed
Manual audit: no serious problem	49 / 59 (83.1%); 10 serious
Guarded pipeline, guard v1(new Gemini calls)	54 shown, 0 fallback, 5 no LLM (low confidence)
Manual read of the 54 shown	0 serious; 10 minor
Guard v2 on the same outputs	44 shown, 10 fall back to rule text, 5 no LLM

The automatic checker passed 54 of 59 rewordings, and after its false alarms were fixed it passed all 59. The manual audit still found 10 with serious problems that the checker could not see. In three cases the LLM named a sign the rules had deliberately withheld, for example rewording an uncertain "angry" attempt as "stomach". In six cases it dropped the "not fully certain" hedge (bread, cough, open, sorry, tea, yes). In one case it read the sign name "weak" as a judgement of quality. The root cause was the prompt, which gave Gemini the predicted sign and the body-part list even when the rule message had left them out.

After the redesign, low-confidence cases never reached Gemini, and Gemini only saw the rule text. On a fresh run none of the serious problems recurred, with 0 serious problems in the 54 rewordings shown. The manual read then found a new, minor issue. Eight rewordings dropped a detail the learner needs, for example "left wrist" became "wrist". Guard v2 added a check that every side and body part is kept. On the same outputs it sent these 8, plus 2 harmless cases, back to the rule text. Two unflagged wording issues remain: one reply implied a cause ("recognised because your execution closely matched"), and one left out an attempt count. Guard v2 was refined after reading these outputs, and it has not yet been run on a fresh set of Gemini replies.

6.2 Product and Technical Results

6.2.1 Speed

Table 6.6 reports inference time over five Colab runs.

Table 6.6: Inference time in milliseconds (five runs)

Set-up	ms per prediction or attempt (runs 1 to 5)
Single TCN prototype	24.0 / 26.0 / 25.8 / 19.4 / 28.5
Ensemble, two separate predict calls	146.5 / 170.4 / 204.9 / 133.9 / 171.4
Ensemble, single file (used in the app)	74.1 / 72.5 / 71.9 / 51.9 / 96.6
Full attempt: recognition + DTW + feedback rules	133.1 / 87.8 / 81.8 / 73.2 / 144.2

The single-file ensemble takes 52 to 97 ms per prediction, about half the time of two separate calls and roughly 30 to 70 ms more than the single TCN. A full attempt, excluding MediaPipe and Gemini, takes 73 to 144 ms. MediaPipe extraction and the

network call to Gemini take far longer than the model, so the move to the ensemble causes no delay a learner would notice. NFR2 is met.

6.2.2 Requirement coverage

All thirteen functional requirements in Table 4.1 are implemented in the product, and Table 5.2 maps each test case to the requirements it covers. FR2 is met by the 2D guide and reference video rather than the 3D avatar first proposed. FR8 is met by the guarded rewording step. Table 6.7 summarises the non-functional requirements.

Table 6.7: Status of the non-functional requirements

ID	Quality	Evidence	Status
NFR1	Usability	Clear screens, camera fallback, live demonstrations; caregiver SUS study prepared	Partly shown; formal test next
NFR2	Performance	73 to 144 ms per attempt (model path)	Met
NFR3	Reliability	Rule text shown on any LLM failure (T11)	Met
NFR4	Correctness of feedback	0 serious problems in 54 shown rewordings; guard v2	Met, with the caveats in Section 6.1.6
NFR5	Security	Token checked on every request; ID derived on the server (T4, T5)	Met
NFR6	Privacy	Landmarks only; LLM sees rule text only	Met by design
NFR7	Portability	Browser and standard webcam	Met
NFR8	Maintainability	Product bundle with fingerprints and parity file	Met
NFR9	Scalability	Prefixed tables, reference rows per sign, module boundary	Met for the pilot scale; worker pool needed for many users

6.3 User or Stakeholder Validation

Stakeholders shaped GLOSS at several points. Table 6.8 lists what caregivers, a sign language tutor and reviewers said, mostly during the CCHSS visit, and what changed as a result.

Table 6.8: Stakeholder feedback and resulting changes

Who	What they said	What changed
Caregiver trainees (CCHSS)	"We only remember a few signs after training."	Focus on continued practice; a fixed 59-sign care vocabulary; signs return until they are mastered
Reviewers	"Recognition alone is not enough for learning."	DTW execution scoring and corrective feedback added
Sign language tutor	"Different signing speeds should still be accepted."	Speed-tolerant DTW comparison; vocabulary check that led to 59 signs
Prototype users	"The system should work without complex hardware."	Browser and standard webcam only; MediaPipe landmarks
Research validation	"Limited samples can cause overfitting."	Parameter budget, full merged dataset, validation-only selection
Remark Elder Care chairperson	Caregivers miss early non-verbal symptoms such as those of Parkinson's disease	Parkinson's symptom module added

At the proposal presentation the panel asked how GLOSS differs from a normal sign language tutor and which sign language it uses. The answers became design features. GLOSS separates recognition from execution quality, explains errors and adapts the lesson order, and it uses ASL for the reasons given in Section 1.2. At the second progress presentation the full pipeline was demonstrated live with a webcam, from recording to feedback and mastery update. Signs were recognised and graded in front of the panel.

The demonstration also showed the safety net working. When "pain" was signed it was recognised as "walk" with a confidence of 0.23, and "hurt" was recognised as "pain" with 0.30. Both fall below 0.35, so the tutor asked the learner to repeat and gave no wrong correction.

A formal evaluation with caregivers is the next step. The protocol and instruments are ready (Section 3.3.6 and Appendix A). They measure performance on three care signs before and after practice, the in-app Parkinson's quiz score, and usability with

the SUS. Until that study is complete, claims about learning gains and usability rest on stakeholder feedback and demonstrations, not on measured outcomes.

6.4 Integration Results

GLOSS runs as the "Sign & Vitals" module of CareSense, next to Deterioration Detection, Fall Detection and Voice Log (Figure 5.2). Caregivers log in once and reach every module from the same sidebar. The GLOSS tables live alongside the other components' tables in the same Supabase project without conflicts. One webcam attempt makes several sequential calls to Supabase, for authentication, the session, the sign reference, the attempt record, the mastery update and the next recommendation. This works well for a single user and is the main reason for the worker-pool change planned in Section 5.5.

6.5 Summary of Findings

- RQ1. The validation-selected ensemble of the improved TCN and SPOTER-style models is the best recogniser, with a test macro-F1 of 0.7752 (95% CI 0.712 to 0.802) and accuracy of 0.7859. Its 0.038 gain in test macro-F1 over the best single model is statistically supported (paired bootstrap 95% CI 0.008 to 0.072).
- RQ2. Confidence is a usable decision signal. Confident predictions are 98.4% correct, and asking for a repeat below 0.35 avoids wrong corrections at the cost of some unnecessary repeats.
- RQ3. DTW against a medoid reference grades quality in a speed-tolerant way (a distance change of 0.337 between 0.7 and 1.3 times speed), but only after the thresholds were calibrated with the live evaluation function. DTW alone cannot replace recognition.
- RQ4. An LLM can safely reword feedback only when it sees nothing but the rule text and every reply is checked. Without these safeguards, 10 of 59 rewordings had serious problems that an automatic checker missed.
- RQ5. The tutoring loop extends naturally to Parkinson's motor symptoms through cards, video questions and per-symptom progress, with an explicit observe-and-report message and no diagnostic model.

7 DISCUSSION

7.1 Interpretation of Findings

The headline number, a macro-F1 of 0.7752 on 59 signs, needs to be read together with the confidence bands. A tutor does not have to be right on every attempt. It has to avoid teaching something wrong. Seen this way, the most useful result is not the overall score but the behaviour at the edges. When the ensemble is confident it is almost always right (98.4%), and when it is unsure the product stays quiet and asks for another try. A learner practising "pain" will sometimes be asked to repeat a correct sign. They will not be told that their "pain" was "walk".

The DTW results show both what speed tolerance offers and where it stops. Signing at 0.7 or 1.3 times normal speed barely changed the distance, which is exactly what a beginner needs. But the overlap between same-sign and different-sign distances (50.85%) shows that DTW on its own is a weak judge of identity. The design response, running DTW only after recognition confirms the sign, came from this finding. The threshold bug points to a more general lesson. A calibration is only valid if it uses exactly the same function as the live system, and the only way to be sure is to test the two against each other.

The feedback experiment may be the most transferable result. An automatic checker passed every rewording once its false alarms were fixed, yet a careful human read found 10 serious problems. In other words, the easy metric said the LLM was safe when it was not. The fix did not depend on a better LLM. It depended on giving the LLM less: only the rule text, nothing to infer from, and a check on what came back.

7.2 Comparison with Literature Baselines and Existing Solutions

Direct comparison with published accuracy figures is difficult because vocabularies, data sources and splits differ. The figures below are context, not a ranking.

- Earlier in this project, a CNN+BiLSTM on 66 signs reached 67.18% accuracy. The final ensemble reaches 78.59% accuracy on 59 signs. The vocabularies differ, but the gain also reflects the parameter budget, kinematic features and ensembling.

- Signing for Care reported a median accuracy of 85.7% for five emergency responders, each testing at least five signs with two tries per sign, with a 79-sign recogniser [12]. GLOSS reports 78.59% over all 59 signs on held-out videos, with a confidence interval. The settings are not comparable, but the Signing for Care users' main request, explanations of why a sign was wrong, is what GLOSS adds.
- PopSignAI's LSTM reached 82.9% on an independent test set [42]. It serves a game where the learner chooses from a small set of signs at each step, while GLOSS judges an open attempt against 59 classes.
- On the full WLASL benchmarks, pose-based models such as SPOTER report results for vocabularies of 100 or more signs from a single source [27], [6]. GLOSS mixes two sources with different camera conditions, which makes the task harder in one way and smaller in another.

Against existing products, GLOSS adds three things none of them offered together: an explanation tied to the body part at fault, a refusal to correct when uncertain, and symptom education alongside signs. It also lacks things some of them have, notably a polished 3D avatar (NVIDIA Signs) and a large lesson catalogue (Lingvano).

7.3 Research Contribution

The research contributions are four.

- A validation-only selection protocol for a small sign recognition dataset, with bootstrap intervals and a paired test showing that an ensemble of a TCN and a pose transformer improves on the best single model.
- Evidence that recogniser confidence can act as a decision rule in a tutor, with measured accuracy and coverage in each band.
- A calibrated, speed-tolerant DTW grading step, together with a documented failure mode (mismatched distance functions) and its fix.
- A faithfulness audit of LLM-reworded tutoring feedback, showing that automatic checks missed serious errors, and a guard design that removed them in a rerun.

7.4 Product Contribution

On the product side, GLOSS is a working module inside a real multi-component platform, not a stand-alone demonstration. Caregivers log in once, practise signs with a webcam, receive feedback in about the time it takes to read it, and see their progress over days. The same learner loop serves the Parkinson's module. The product bundle with fingerprints and a parity file makes the link between the thesis results and the running system checkable by anyone.

7.5 Limitations and Threats to Validity

- Signer overlap. The split is stratified by sign, not by signer, so the same people can appear in training and test. Accuracy on new signers, such as real caregivers, is likely lower [53]. A signer-independent split is planned.
- Small test set per sign. Four to ten videos per sign make per-sign F1 values indicative only. The bootstrap interval reports the overall uncertainty.
- Fluent reference signers. The datasets contain fluent adults, not novice caregivers. Beginners make different mistakes, and the thresholds may need adjustment after the caregiver study.
- Faithfulness sample size. The audit covered 59 rewordings, and both the audit checks and the v2 detail check were written after reading the outputs. Guard v2 has not been tested on a fresh set of replies.
- Prototype recogniser in the faithfulness experiment. That experiment used the prototype recogniser. It measures the rewording step, which does not depend on which recogniser produced the rule text.
- Confidence cut-offs. The 0.35 and 0.6 cut-offs were set by design. Tuning them on validation data could improve the trade-off.
- Scope. Avatar guidance (Layer 5) and continuous real-time signing are not implemented. The Parkinson's content is educational, with no classifier.
- User study. GLOSS has not yet been evaluated with caregivers in a formal study. Usability and learning gains are therefore not yet measured.
- Language. GLOSS teaches ASL. Deaf residents in Sri Lanka mostly use Sri Lankan Sign Language, so local use needs localisation.

- Concurrency. The back end blocks during an attempt, which limits the number of simultaneous users until a worker pool is added.

7.6 Implications

For caregiver training schools, GLOSS suggests a practical model: teach signs in class, then give trainees a tutor that keeps them practising afterwards and shows instructors who is falling behind. For care homes, the Parkinson's module offers a low-cost way to make staff more alert to early motor changes without any clinical risk, because it teaches observation, not diagnosis.

For people building AI tutors, two lessons carry over. A tutor should use its own uncertainty and should be designed to say "try again" rather than guess. And when an LLM is added to make feedback friendlier, it should be treated as untrusted. It should receive the minimum input and its output should be checked, with a correct fallback always ready.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

GLOSS helps caregivers keep practising care-related sign language after their training ends and learn to notice the motor signs of Parkinson's disease. It does this with feedback that explains what to fix and never corrects on a guess. Training providers and care homes get a way to keep these skills alive without paying a tutor to sit with every caregiver.

Four customer groups were identified from the industry visits.

Table 8.1: Target users and what GLOSS offers each

User group	Need	What GLOSS offers
Caregiver trainee (pre-departure)	Must show basic communication and care skills to overseas employers	Daily practice on care signs, progress record, certificate in a later version
Care home caregiver	Communicates with residents who cannot hear or speak; notices health changes	Short sessions on a phone or laptop; symptom cards for residents with Parkinson's
Training school coordinator	Sign language is taught briefly and forgotten	Continued practice for each cohort and visibility of weak signs
Care home owner or manager	Quality of care, staff confidence, low training cost	Low monthly fee, no hardware, pilot before buying

8.2 Market and Competitive Position

Sri Lanka is ageing quickly. The 2024 census counted 3,921,124 people aged 60 or over, about 18% of the population [54]. Demand for trained caregivers is rising at home and abroad. Under the bilateral agreement with Israel alone, 2,656 Sri Lankans had been placed in caregiver jobs by January 2026, with about 1,000 more planned for 2026 [55]. Candidates need a caregiver course registered with the Sri Lanka Bureau of Foreign Employment together with an NVQ Level III certificate, or at least 11 months of nursing experience [55]. This creates a steady flow of trainees through training schools such as CCHSS, which is the natural sales channel for GLOSS.

The need for symptom awareness grows with the same ageing trend. About 11.9 million people worldwide were living with Parkinson's disease in 2021, and the number is projected to keep rising as populations age [5].

Table 8.2 places GLOSS against the main alternatives.

Table 8.2: Competitive position

Product	Price	Strength	Gap that GLOSS fills
Lingvano	USD 19.99 per month (about Rs 6,600) [41]	Polished general ASL course	No care vocabulary; no symptom content
NVIDIA Signs	Free [15]	Webcam practice, 3D avatar	General vocabulary; no caregiver focus or progress for institutions
Signing for Care	Research prototype [12]	Emergency medical vocabulary	Users asked for explanations; not available as a product
Classroom training only	Course fee	Human teacher	No practice after the course; skills fade
GLOSS	From Rs 990 per month (individual)	Care signs, explained feedback, Parkinson's module, institution view	Smaller vocabulary; ASL only for now

GLOSS competes on focus rather than breadth. It is cheaper than a general app and teaches exactly what a caregiver needs. It can also be sold as part of CareSense to care homes that want the other three modules.

8.3 Business or Sustainability Model

8.3.1 Revenue model and pricing

GLOSS is offered as software as a service, either alone or as the Sign & Vitals module of CareSense. Table 8.3 shows the proposed tiers. Annual plans are charged at ten months' price.

Table 8.3: Proposed pricing (Sri Lankan rupees)

Tier	Who it is for	Price
Individual	One caregiver or trainee	Rs 990 per month, or Rs 4,900 for a six-month pre-departure pass
Starter	Care home with up to 10 caregivers	Rs 12,500 per month
Standard	Up to 30 caregivers, supervisor progress view	Rs 25,000 per month

Tier	Who it is for	Price
Institution	Training school, up to 100 trainees, cohort reports and certificates	Rs 45,000 per month
Onboarding	Set-up, staff training and device check (one time)	Rs 40,000

The individual price is about one sixth of a general ASL app. The institutional tiers are priced below the cost of one part-time instructor.

8.3.2 Cost model

Table 8.4 lists the monthly running costs. Dollar prices were converted at Rs 330.61 per USD, the rate on 8 October 2026 [56].

Table 8.4: Monthly running cost of the platform

Item	Basis	Rs per month
Supabase Pro (database, auth)	Actual bill during development [57]	9,000
Back-end hosting, 1 CPU and 2 GB RAM	USD 25 [58]	8,300
Gemini rewording	About 400 input and 80 output tokens per call [59]; about 10,000 attempts	1,000
Cloudflare R2 video storage	Estimate, USD 20	6,600
Domain, e-mail and other	Estimate	1,500
Platform total		26,400
Part-time maintenance and support	Estimate	60,000
Fixed cost to recover		about 86,400

The variable cost per attempt is very small. A Gemini rewording costs roughly a tenth of a rupee, and the low-confidence attempts never call Gemini at all. Most costs are fixed, so margins improve quickly as customers are added.

8.3.3 Break-even

One training school (Rs 45,000), two Starter care homes (Rs 25,000) and twenty individual subscribers (Rs 19,800) bring in about Rs 89,800 a month, which covers the fixed cost of about Rs 86,400. Three institutional customers and a modest number of individual subscribers are therefore enough to sustain the product. Onboarding fees cover the extra cost of setting up each new site.

8.3.4 Go-to-market plan

1. Pilot. Offer a free three-month full deployment of CareSense, including GLOSS, at a care home in Sri Lanka. Measure usage, sign learning and Parkinson's knowledge with the evaluation instruments in Appendix A, and ask for a letter of intent at the end.
2. Training-school channel. Offer GLOSS to caregiver training schools as a practice tool bundled with their caregiver course, priced per cohort.
3. Expansion. Add a certificate of completion for pre-departure trainees and sell the individual pass through training schools.

8.3.5 Sustainability

The product is cheap to run because the heavy parts (MediaPipe and the recogniser) run on a modest server, and the LLM step is optional. If Gemini prices rise or the service becomes unavailable, GLOSS keeps working with the rule-based text. Content can grow by adding reference sequences for new signs, without retraining the whole system for small additions to the guide.

8.4 Scalability Intellectual Property and Compliance

8.4.1 Scalability

New signs need reference recordings and a model update. New learners need only accounts. New care homes need no new infrastructure because tables are shared and every row belongs to a caregiver. Before larger deployments, a worker pool is needed so that one attempt does not block others. Beyond about a hundred active users, MediaPipe extraction could move to the browser so the server only receives landmarks. That would cut both server load and the amount of data sent.

ASL was chosen because it is the most widely resourced sign language and suits curricula aimed at international caregiving. The pipeline itself does not depend on a particular sign language. Adding Sri Lankan Sign Language, or another national sign language, mainly requires new reference recordings and training data, which makes localisation a credible next market.

8.4.2 Intellectual property

The GLOSS source code, the interface design and the curated content (care vocabulary, symptom cards and quiz bank) are protected by copyright. The product name and the CareSense brand can be registered as trademarks. The calibrated thresholds, the feedback rules and the guard design are best kept as know-how rather than patented, because patents on software methods are hard to obtain and easy to work around. Third-party components are used under their open-source licences, for example MediaPipe and TensorFlow under the Apache 2.0 licence and React and FastAPI under the MIT licence.

The training datasets are the main IP risk. WLASL and ASL Citizen were released for research, and their terms must be confirmed before paid use. The safest commercial path is to record a licensed reference set with Deaf signers, which would also improve quality.

8.4.3 Compliance

GLOSS processes personal data in the form of names, accounts and learning records. It does not store video. Sri Lanka's Personal Data Protection Act, No. 9 of 2022, applies, and its key provisions come into operation on 1 January 2027 [60], [61]. The design already follows its main principles: data minimisation, purpose limitation and secure processing. The Parkinson's module makes no diagnostic claim and analyses no patient data, which keeps GLOSS outside medical device regulation. Any future feature that analyses a resident's movements would need clinical validation and regulatory review first.

8.5 Contribution to the Sustainable Development Goals

GLOSS contributes to three of the United Nations Sustainable Development Goals [62].

- SDG 3, Good Health and Well-being. Better communication with deaf and non-verbal residents, and earlier reporting of motor symptoms, support safer care for older adults.

- SDG 4, Quality Education. Caregivers gain affordable, self-paced practice with explained feedback, which keeps skills alive after formal training.
- SDG 10, Reduced Inequalities. Deaf and hard-of-hearing people get caregivers who can communicate basic needs in their language, which reduces a long-standing gap in access to care.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis set out to give caregivers a way to keep learning the non-verbal signals of the people they look after, both the signs a person makes with their hands and the signs their body shows. GLOSS does this in one tutor. A caregiver records a care sign with a webcam. The system recognises it with an ensemble of a TCN and a pose transformer, grades how well it was made with speed-tolerant DTW, and explains what to change in plain words. The next sign is chosen from the learner's own practice history, and a sign counts as mastered only after three consecutive strong attempts. The same loop teaches six motor symptoms of Parkinson's disease, always with the message to observe and report rather than diagnose.

The results support the main design choices. The validation-selected ensemble outperformed the best single model, its confidence proved reliable enough to decide between correcting and asking for a repeat, DTW grading became informative once its thresholds were calibrated with the live distance function, and the guard removed the serious faithfulness problems found in the LLM rewording.

9.2 Achievement of Objectives

Table 9.1: Achievement of the research objectives

ID	Objective	Status	Evidence
O1	Care vocabulary and clean landmark dataset	Achieved	59 signs checked with a sign language tutor; 2,364 sequences; reproducible rerun (Sections 3.2, 3.7)
O2	Validation-selected recogniser with confidence intervals	Achieved	Ensemble macro-F1 0.7752, CI 0.712 to 0.802; paired gain 0.038 (Section 6.1)
O3	Speed-tolerant execution evaluation with error location	Achieved	DTW with v3 thresholds; per-landmark error; speed test (Section 6.1.5)
O4	Interpretable feedback, faithful LLM wording, visual guide	Achieved, with a change	Rule engine and guard v2; 2D guide and reference video instead of the 3D avatar (Sections 4.3, 6.1.6)
O5	Mastery tracking and adaptive lessons	Achieved	Mastery engine, lesson selector, progress screens (Sections 4.4.6, 5.2.3)

ID	Objective	Status	Evidence
O6	Non-diagnostic Parkinson's module	Achieved	Symptom cards, video quiz, per-symptom progress (Sections 4.4.8, 5.2.4)
O7	Integration and evaluation	Largely achieved	Running in CareSense; accuracy, reliability and speed evaluated; caregiver usability study prepared (Chapter 6)

9.3 Individual Research and Product Contributions

My research contributions are a careful selection and evaluation protocol for a small sign recognition dataset, evidence that confidence bands can steer tutoring decisions, a calibrated speed-tolerant grading step with a documented failure mode, and an audit-driven guard that makes LLM-reworded feedback faithful. My product contribution is GLOSS itself, a working module of CareSense with a FastAPI back end, a Supabase data model, React screens for signs and symptoms, and a fingerprinted product bundle that links the running system to the reported results.

9.4 Future Directions

- Run the caregiver evaluation (Appendix A) with care home staff and trainees, and measure sign learning, Parkinson's symptom knowledge and SUS usability. Follow up after two weeks to measure retention.
- Build a signer-independent test set, including recordings from Sri Lankan caregivers, to measure accuracy on new people.
- Tune the confidence cut-offs on validation data and recalibrate the DTW thresholds with novice attempts.
- Run guard v2 on a fresh set of Gemini replies, and repeat the faithfulness audit with the final recogniser.
- Improve the weakest signs, especially hurt and pain, with more training videos and targeted features for their small differences in movement.
- Localise GLOSS to Sri Lankan Sign Language and offer instructions in Sinhala and Tamil.
- Complete Layer 5 with a rigged avatar that retargets the reference landmarks onto a natural body, and move towards continuous, real-time feedback.

- Extend the symptom module to other conditions common in older adults, such as dementia and stroke, keeping the same non-diagnostic, observe-and-report design.
- Move blocking work to a worker pool before a multi-site pilot.

REFERENCES

[1] K. J. Muir, T. N. Wanchek, J. M. Lobo, and J. Keim-Malpass, "Evaluating the costs of nurse burnout-attributed turnover: A Markov modeling approach," J. Patient Saf., vol. 18, no. 4, pp. 351–357, Jun. 2022, doi: 10.1097/PTS.0000000000000920.

[2] Office of the U.S. Surgeon General, "Addressing health worker burnout: The U.S. Surgeon General's advisory on building a thriving health workforce," U.S. Dept. of Health and Human Services, Washington, DC, USA, 2022.

[3] Centers for Disease Control and Prevention, "Older adult falls data," National Center for Injury Prevention and Control. Accessed: Oct. 10, 2026. [Online]. Available: https://www.cdc.gov/falls/data-research/

[4] World Health Organization, World Report on Hearing. Geneva, Switzerland: WHO, 2021.

[5] D. Su et al., "Projections for prevalence of Parkinson's disease and its driving factors in 195 countries and territories to 2050: Modelling study of Global Burden of Disease Study 2021," BMJ, vol. 388, Art. no. e080952, 2025, doi: 10.1136/bmj-2024-080952.

[6] D. Li, C. Rodriguez, X. Yu, and H. Li, "Word-level deep sign language recognition from video: A new large-scale dataset and methods comparison," in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV), 2020, pp. 1448–1458, doi: 10.1109/WACV45572.2020.9093512.

[7] A. Desai et al., "ASL Citizen: A community-sourced dataset for advancing isolated sign language recognition," in Adv. Neural Inf. Process. Syst. (NeurIPS), Datasets and Benchmarks Track, vol. 36, 2023, pp. 76893–76907.

[8] O. Qadhi, "Sign language use in healthcare: Professionals' insight," PeerJ, vol. 13, Art. no. e19446, 2025, doi: 10.7717/peerj.19446.

[9] S. A. Jacob, U. D. Palanisamy, J. Napier, D. Verstegen, A. Dhanoa, and E. Y.-C. Chong, "Health care needs of deaf signers: The case for culturally competent health care providers," Acad. Med., vol. 97, no. 3, pp. 335–340, 2022, doi: 10.1097/ACM.0000000000004181.

[10] M. Abou-Abdallah and A. Lamyman, "Exploring communication difficulties with deaf patients," Clin. Med., vol. 21, no. 4, pp. e380–e383, Jul. 2021, doi: 10.7861/clinmed.2021-0111.

[11] N. J. Cepeda, H. Pashler, E. Vul, J. T. Wixted, and D. Rohrer, "Distributed practice in verbal recall tasks: A review and quantitative synthesis," Psychol. Bull., vol. 132, no. 3, pp. 354–380, 2006, doi: 10.1037/0033-2909.132.3.354.

[12] C. Kim, C. McLaren, M. Krishnan, N. Modayur, and S. Hassan, "Signing for Care: A demo and initial evaluation of an American Sign Language learning tool for emergency medical service providers," in Proc. 27th Int. ACM SIGACCESS Conf. Comput. Accessibility (ASSETS), Denver, CO, USA, 2025, doi: 10.1145/3663547.3759746.

[13] M. Huenerfauth, E. Gale, B. Penly, S. Pillutla, M. Willard, and D. Hariharan, "Evaluation of language feedback methods for student videos of American Sign Language," ACM Trans. Access. Comput., vol. 10, no. 1, Art. no. 2, Apr. 2017, doi: 10.1145/3046788.

[14] R. B. Postuma et al., "MDS clinical diagnostic criteria for Parkinson's disease," Mov. Disord., vol. 30, no. 12, pp. 1591–1601, 2015, doi: 10.1002/mds.26424.

[15] NVIDIA, "Signs: Learn and contribute to American Sign Language with AI," 2025. [Online]. Available: https://signs-ai.com/

[16] A. Battisti and S. Ebling, "Automatic annotation elaboration as feedback to sign language learners," in Proc. 18th Linguistic Annotation Workshop (LAW-XVIII), St. Julians, Malta, 2024, pp. 46–60, doi: 10.18653/v1/2024.law-1.5.

[17] J. Hattie and H. Timperley, "The power of feedback," Rev. Educ. Res., vol. 77, no. 1, pp. 81–112, 2007, doi: 10.3102/003465430298487.

[18] B. S. Bloom, "The 2 sigma problem: The search for methods of group instruction as effective as one-to-one tutoring," Educ. Researcher, vol. 13, no. 6, pp. 4–16, 1984, doi: 10.3102/0013189X013006004.

[19] K. VanLehn, "The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems," Educ. Psychol., vol. 46, no. 4, pp. 197–221, 2011, doi: 10.1080/00461520.2011.611369.

[20] A. T. Corbett and J. R. Anderson, "Knowledge tracing: Modeling the acquisition of procedural knowledge," User Model. User-Adapt. Interact., vol. 4, no. 4, pp. 253–278, 1994, doi: 10.1007/BF01099821.

[21] B. R. Bloem, M. S. Okun, and C. Klein, "Parkinson's disease," Lancet, vol. 397, no. 10291, pp. 2284–2303, 2021, doi: 10.1016/S0140-6736(21)00218-X.

[22] C. Lugaresi et al., "MediaPipe: A framework for building perception pipelines," arXiv preprint arXiv:1906.08172, 2019.

[23] V. Bazarevsky, I. Grishchenko, K. Raveendran, T. Zhu, F. Zhang, and M. Grundmann, "BlazePose: On-device real-time body pose tracking," arXiv preprint arXiv:2006.10204, 2020.

[24] I. Grishchenko and V. Bazarevsky, "MediaPipe Holistic: Simultaneous face, hand and pose prediction, on device," Google Research Blog, Dec. 2020. [Online]. Available: https://research.google/blog/mediapipe-holistic-simultaneous-face-hand-and-pose-prediction-on-device/

[25] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," arXiv preprint arXiv:1803.01271, 2018.

[26] A. Vaswani et al., "Attention is all you need," in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017, pp. 5998–6008.

[27] M. Bohacek and M. Hruz, "Sign pose-based transformer for word-level sign language recognition," in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW), 2022, pp. 182–191, doi: 10.1109/WACVW54805.2022.00024.

[28] S. Yan, Y. Xiong, and D. Lin, "Spatial temporal graph convolutional networks for skeleton-based action recognition," in Proc. 32nd AAAI Conf. Artif. Intell., 2018, pp. 7444–7452, doi: 10.1609/aaai.v32i1.12328.

[29] S. Jiang, B. Sun, L. Wang, Y. Bai, K. Li, and Y. Fu, "Skeleton aware multi-modal sign language recognition," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW), 2021, pp. 3408–3418, doi: 10.1109/CVPRW53098.2021.00380.

[30] T. G. Dietterich, "Ensemble methods in machine learning," in Multiple Classifier Systems (Lecture Notes in Computer Science, vol. 1857). Berlin, Germany: Springer, 2000, pp. 1–15, doi: 10.1007/3-540-45014-9_1.

[31] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017, pp. 6402–6413.

[32] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in Proc. 34th Int. Conf. Mach. Learn. (ICML), 2017, pp. 1321–1330.

[33] H. Sakoe and S. Chiba, "Dynamic programming algorithm optimization for spoken word recognition," IEEE Trans. Acoust., Speech, Signal Process., vol. 26, no. 1, pp. 43–49, 1978, doi: 10.1109/TASSP.1978.1163055.

[34] T. Giorgino, "Computing and visualizing dynamic time warping alignments in R: The dtw package," J. Stat. Softw., vol. 31, no. 7, pp. 1–24, 2009, doi: 10.18637/jss.v031.i07.

[35] H. Wen, Y. Xu, L. Li, X. Ru, X. Wang, and Z. Wu, "Enhancing sign language teaching: A mixed reality approach for immersive learning and multi-dimensional feedback," arXiv preprint arXiv:2404.10490, 2024.

[36] Y. Zhang, Y. Min, and X. Chen, "Teaching Chinese sign language with a smartphone," Virtual Real. Intell. Hardw., vol. 3, no. 3, pp. 248–260, Jun. 2021, doi: 10.1016/j.vrih.2021.05.004.

[37] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, "On faithfulness and factuality in abstractive summarization," in Proc. 58th Annu. Meeting Assoc. Comput. Linguistics (ACL), 2020, pp. 1906–1919, doi: 10.18653/v1/2020.acl-main.173.

[38] Z. Ji et al., "Survey of hallucination in natural language generation," ACM Comput. Surv., vol. 55, no. 12, Art. no. 248, 2023, doi: 10.1145/3571730.

[39] K. Chemnad and A. Othman, "Perception and monitoring of sign language acquisition for avatar technologies: A rapid focused review (2020–2025)," Multimodal Technol. Interact., vol. 9, no. 8, Art. no. 82, 2025, doi: 10.3390/mti9080082.

[40] L. Dong, L. Chaudhary, F. Xu, X. Wang, M. Lary, and I. Nwogu, "SignAvatar: Sign language 3D motion reconstruction and generation," arXiv preprint arXiv:2405.07974, 2024.

[41] Lingvano GmbH, "Lingvano - Learn Sign Language," Apple App Store listing, accessed Oct. 2026. [Online]. Available: https://apps.apple.com/app/id1547252782

[42] D. Martin et al., "PopSignAI: Towards using sign language recognition games to improve American Sign Language learning in novice signers," in Proc. 31st Int. Conf. Intell. User Interfaces (IUI), 2026, pp. 1298–1321, doi: 10.1145/3742413.3789164.

[43] X. Liu, H.-L. Cheng, G. Chastel, M. Chastel, and Z. Bai, "CoSignPlay: A collaborative approach to learning non-manual signs in ASL for hearing families with deaf children," in Proc. 27th Int. ACM SIGACCESS Conf. Comput. Accessibility (ASSETS), 2025, doi: 10.1145/3663547.3746382.

[44] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," MIS Quart., vol. 28, no. 1, pp. 75–105, 2004, doi: 10.2307/25148625.

[45] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A design science research methodology for information systems research," J. Manage. Inf. Syst., vol. 24, no. 3, pp. 45–77, 2007, doi: 10.2753/MIS0742-1222240302.

[46] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), 2017, pp. 2999–3007, doi: 10.1109/ICCV.2017.324.

[47] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the Inception architecture for computer vision," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 2818–2826, doi: 10.1109/CVPR.2016.308.

[48] Gemini Team, Google, "Gemini: A family of highly capable multimodal models," arXiv preprint arXiv:2312.11805, 2023.

[49] J. Brooke, "SUS: A quick and dirty usability scale," in Usability Evaluation in Industry, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[50] B. Efron and R. J. Tibshirani, An Introduction to the Bootstrap. New York, NY, USA: Chapman & Hall, 1993.

[51] A. Bangor, P. T. Kortum, and J. T. Miller, "An empirical evaluation of the System Usability Scale," Int. J. Hum.-Comput. Interact., vol. 24, no. 6, pp. 574–594, 2008, doi: 10.1080/10447310802205776.

[52] Systems and Software Engineering - Systems and Software Quality Requirements and Evaluation (SQuaRE) - System and Software Quality Models, ISO/IEC 25010:2011, 2011.

[53] A. Torralba and A. A. Efros, "Unbiased look at dataset bias," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2011, pp. 1521–1528, doi: 10.1109/CVPR.2011.5995347.

[54] Press Trust of India, "Sri Lanka ageing fast; elderly population triples in 40 years," ETV Bharat, Apr. 11, 2026, reporting the Department of Census and Statistics, Census of Population and Housing 2024. [Online]. Available: https://www.etvbharat.com/en/international/sri-lanka-ageing-fast-elderly-population-triples-in-40-yrs-enn26041104088

[55] Ada Derana, "Sri Lanka to send 1,000 caregivers to Israel in 2026," Jan. 20, 2026. [Online]. Available: https://www.adaderana.lk/news/117373/sri-lanka-to-send-1000-caregivers-to-israel-in-2026

[56] Exchange-Rates.org, "USD to LKR exchange rate," Oct. 8, 2026. [Online]. Available: https://www.exchange-rates.org/converter/usd-lkr

[57] MakerKit, "Supabase pricing in 2026: What you'll actually pay," 2026. [Online]. Available: https://makerkit.dev/blog/saas/supabase-pricing

[58] Render, "Pricing," accessed Oct. 2026. [Online]. Available: https://render.com/pricing

[59] Price Per Token, "Google Gemini 2.5 Flash pricing," accessed Oct. 2026. [Online]. Available: https://pricepertoken.com/pricing-page/model/google-gemini-2.5-flash

[60] Parliament of the Democratic Socialist Republic of Sri Lanka, Personal Data Protection Act, No. 9 of 2022. Colombo, Sri Lanka, 2022.

[61] Government of Sri Lanka, "Personal Data Protection Act, No. 9 of 2022: Order under Section 1(3)," Gazette Extraordinary No. 2498/16, Jul. 22, 2026. [Online]. Available: https://documents.gov.lk/view/egz/2026/7/2498-16_E.pdf

[62] United Nations General Assembly, "Transforming our world: The 2030 Agenda for Sustainable Development," Resolution A/RES/70/1, 2015.

GLOSSARY

Term	Definition
Bootstrap confidence interval	A range for a metric obtained by resampling the test set many times with replacement and taking the middle 95% of the resampled scores.
Bradykinesia	Slowness of movement, with movements becoming smaller or more effortful; a core motor sign of Parkinson's disease.
Confidence band	One of three ranges of recogniser confidence (below 0.35, 0.35 to 0.6, 0.6 and above) that decide how the tutor responds.
Ensemble	A model whose prediction is the average of the predictions of several separately trained models.
Faithfulness	The property of a generated text that says nothing beyond what its source supports.
Freezing of gait	A sudden, brief inability to move the feet forward, often at doorways, turns or the first step.
Guard	The GLOSS module that decides whether an LLM rewording may be shown or the rule text must be used instead.
Landmark	A tracked body point, such as a fingertip or wrist, with x, y and z coordinates per video frame.
Macro-F1	The average of the F1 scores of all classes, giving each sign equal weight.
Medoid	The example of a class with the smallest total distance to the other examples of that class.
Parity test	A check that the deployed system reproduces stored outputs of the reported model exactly.

APPENDICES

Appendix A Research Instruments

The caregiver evaluation is designed to take 10 to 15 minutes per participant. Table A.1 lists its steps. A separate organisation questionnaire and a letter of intent template are provided for care home and training school managers and are not part of the timed session.

Table A.1: Steps of the caregiver evaluation session

Step	Activity	Minutes	Completed by
1	Consent and three background questions (role, experience, prior sign knowledge)	2	Participant
2	Pre-test: attempt help, water and sick without the guide, scored 0 to 2	1	Researcher
3	Practice the three signs with GLOSS (2D guide, feedback, up to three attempts each)	5	Participant
4	Post-test: attempt the three signs again through GLOSS	1	Researcher
5	Explore the Parkinson's symptom cards and complete one in-app quiz	3	Participant
6	System Usability Scale, five GLOSS statements, two cost questions and one comment	3	Participant

The SUS items follow Brooke's wording [49]. Odd items are scored as the answer minus one, even items as five minus the answer, and the sum is multiplied by 2.5 to give a score out of 100.

Table A.2: GLOSS-specific statements (five-point agreement scale)

No.	Statement
1	The feedback told me clearly what to change in my sign.
2	When the system asked me to repeat a sign, I understood why.
3	The 2D sign guide helped me learn how the sign moves.
4	The Parkinson's symptom cards and example videos helped me recognise the symptoms.
5	I would keep using GLOSS to practise after finishing a training course.

The two cost questions ask who should pay for a tool such as GLOSS (employer, training school, the caregiver, or free provision) and how much the caregiver would pay personally each month (from nothing to more than Rs 2,000).

Appendix B Disclosure of AI Assistance

I hereby declare that:

1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. I agree to use AI tools responsibly, ethically and transparently.
4. I will clearly disclose any use of AI tools in my submissions.
5. I understand that I bear full responsibility and accountability for any content, code, model or output generated using AI tools.
6. I confirm that I will be able to explain, justify and defend all aspects of my work during evaluations.
7. I understand that failure to comply with this policy may result in academic penalties.

In line with this declaration, I used AI tools as support tools during the preparation of this thesis. They helped me to organise the structure against the faculty template, to draft and edit text from my own project notes, to format the document, to check the details of the cited references and to review and debug code. The research design, the experiments, the interpretation of the results and the conclusions are my own. I reviewed the text before submission, and I take full responsibility for its content, its claims and its citations.

Appendix C Product Bundle

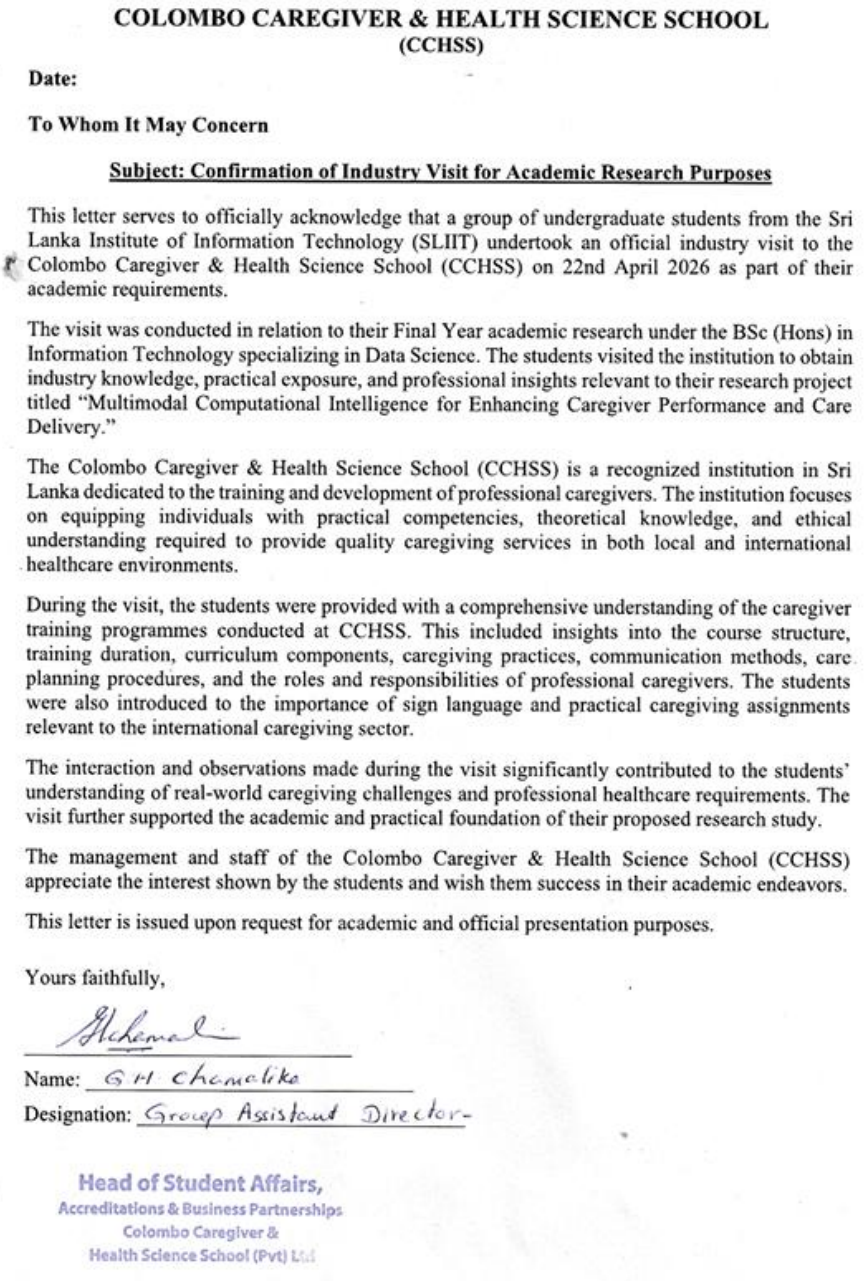
The files below make up the release version of the GLOSS model. Each fingerprint lets a reviewer confirm that a deployed system uses exactly these files.

Table C.1: Product bundle files and fingerprints

File	Fingerprint	Role
GLOSS_ensemble.keras	87693587f51c	Single-file ensemble loaded by the app
TCN_phase1.keras	09830d629d4c	Ensemble member (traceability)
SPOTER-style_phase1.keras	34d8a2e874fd	Ensemble member (traceability)
feature_scaler.npz	48e3bcee6d07	Phase-1 scaler for the 441 features
label_encoder.pkl	361d4640faab	Class label mapping
class_names.json	c5bf6ab3cbff	Sign names
class_thresholds.json (v3)	8797064a2ccf	DTW quality thresholds
reference_exemplars.npz	8f867dd57281	Medoid reference sequences
landmark_names.json	17a5aa3c0d0d	Landmark order
parity_check.npz	a1964400db9f	Ten test sequences and expected outputs
X_test / y_test	a4cda6e8def3 / b2464fb59a02	Frozen test set

Appendix D Industry Visit Confirmation

Letters confirming the industry visits to the Colombo Caregiver and Health Science School (22 April 2026) and Remark Elder Care (Pvt) Ltd (7 April 2026) are reproduced below.



COLOMBO CAREGIVER & HEALTH SCIENCE SCHOOL
(CCHSS)

Date:

To Whom It May Concern

Subject: Confirmation of Industry Visit for Academic Research Purposes

This letter serves to officially acknowledge that a group of undergraduate students from the Sri Lanka Institute of Information Technology (SLIIT) undertook an official industry visit to the Colombo Caregiver & Health Science School (CCHSS) on 22nd April 2026 as part of their academic requirements.

The visit was conducted in relation to their Final Year academic research under the BSc (Hons) in Information Technology specializing in Data Science. The students visited the institution to obtain industry knowledge, practical exposure, and professional insights relevant to their research project titled "Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery."

The Colombo Caregiver & Health Science School (CCHSS) is a recognized institution in Sri Lanka dedicated to the training and development of professional caregivers. The institution focuses on equipping individuals with practical competencies, theoretical knowledge, and ethical understanding required to provide quality caregiving services in both local and international healthcare environments.

During the visit, the students were provided with a comprehensive understanding of the caregiver training programmes conducted at CCHSS. This included insights into the course structure, training duration, curriculum components, caregiving practices, communication methods, care planning procedures, and the roles and responsibilities of professional caregivers. The students were also introduced to the importance of sign language and practical caregiving assignments relevant to the international caregiving sector.

The interaction and observations made during the visit significantly contributed to the students' understanding of real-world caregiving challenges and professional healthcare requirements. The visit further supported the academic and practical foundation of their proposed research study.

The management and staff of the Colombo Caregiver & Health Science School (CCHSS) appreciate the interest shown by the students and wish them success in their academic endeavors.

This letter is issued upon request for academic and official presentation purposes.

Yours faithfully,

Name: G H Chamalika

Designation: Group Assistant Director-

Head of Student Affairs,
Accreditations & Business Partnerships
Colombo Caregiver &
Health Science School (Pvt) Ltd

Figure D.1: Visit confirmation letter from the Colombo Caregiver and Health Science School

REMARK ELDER CARE (PVT) LIMITED

Date:

To Whom It May Concern,

Subject: Confirmation of Industry Visit for Academic Research Purposes

This letter serves to officially confirm that a group of undergraduate students from the Sri Lanka Institute of Information Technology (SLIIT) undertook an official industry visit to Remark Elder Care (Pvt) Limited on 7th April 2026 as part of their academic requirements.

The visit was conducted in relation to their Final Year academic research under the BSc (Hons) in Information Technology specializing in Data Science. The students visited the institution to obtain industry knowledge, practical exposure, and professional insights relevant to their research project titled *"Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery."*

During the visit, the students were provided with valuable insights into the eldercare and caregiving sector, including caregiving operations, service delivery practices, caregiver responsibilities, and the real-world challenges faced within care home environments. The visit also highlighted the gap between formal caregiver training and practical application, enabling the students to better understand the competencies, working conditions, and professional requirements necessary for effective and quality caregiving services in Sri Lanka. The knowledge and observations gained during the visit significantly contributed to the practical and academic foundation of their research study.

The management and staff of Remark Elder Care (Pvt) Limited appreciate the interest shown by the students and wish them success in their academic endeavors.

This letter is issued upon request for academic and official presentation purposes.

Yours faithfully,

Name: Kavishmila Sathsarani

Designation: Manager

REMARK ELDER CARE (PVT) LTD PV 00293928

Figure D.2: Visit confirmation letter from Remark Elder Care